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CATIONIC STEROIDAL ANTIMICROBIAL SALTS

Abstract

Disclosed herein are acid addition salts of cationic steroidal antimicrobials (“CSAs” or “ceragenins”) and methods of making the same. Particularly advantageous salt forms are identified, such as 1,5-naphthalenedisulfonic acid addition salts and sulfate addition salts. The acid addition salts may be formulated for treating subjects with ailments responsive to CSAs, including but not limited to treating bacterial infections. Accordingly, some embodiments include formulations and methods of administering acid addition salts of CSAs.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation-in-part of U.S. patent application Ser. No. 15/135,928, filed Apr. 22, 2016, which claims the benefit of U.S. Provisional Patent Application No. 62/151,019, filed Apr. 22, 2015, U.S. Provisional Patent Application No. 62/165,013, filed May 21, 2015, and U.S. Provisional Patent Application No. 62/191,916, filed Jul. 13, 2015. The disclosures of the foregoing applications are incorporated by reference in their entirety.

BACKGROUND

1. Field

[0002] The present application relates to the fields of pharmaceutical chemistry, biochemistry, and medicine. In particular, the present application relates to acid addition salts of cationic steroidal antimicrobials (“CSAs” or “ceragenins”).

2. Related Technology

[0003] Endogenous antimicrobial peptides, such as the human cathelicidin LL-37, play key roles in innate immunity. LL-37 is found in airway mucus and is believed to be important in controlling bacterial growth in the lung. Antimicrobial peptides are found in organisms ranging from mammals to amphibians to insects to plants. The ubiquity of antimicrobial peptides has been used as evidence that these compounds do not readily engender bacterial resistance. In addition, considering the varied sequences of antimicrobial peptides among diverse organisms, it is apparent that they have evolved independently multiple times. Thus, antimicrobial peptides appear to be one of “Nature's” primary means of controlling bacterial growth. However, clinical use of antimicrobial peptides presents significant issues including the relatively high cost of producing peptide-based therapeutics, the susceptibility of peptides to proteases generated by the host and by bacterial pathogens, and deactivation of antimicrobial peptides by proteins and DNA in lung mucosa.

[0004] An attractive means of harnessing the antibacterial activities of antimicrobial peptides without the issues delineated above is to develop non-peptide mimics of antimicrobial peptides that display the same broad-spectrum antibacterial activity utilizing the same mechanism of action. Non-peptide mimics would offer lower-cost synthesis and potentially increased stability to proteolytic degradation. In addition, control of water solubility and charge density may be used to control association with proteins and DNA in lung mucosa.

[0005] With over 1,600 examples of antimicrobial peptides known, it is possible to categorize the structural features common to them. While the primary sequences of these peptides vary substantially, morphologies adopted by a vast majority are similar. Those that adopt alpha helix conformations juxtapose hydrophobic side chains on one face of the helix with cationic (positively charged) side chains on the opposite side. As similar morphology is found in antimicrobial peptides that form beta sheet structures: hydrophobic side chains on one face of the sheet and cationic side chains on the other.

[0006] Small molecule, non-peptide mimics of antimicrobial peptides, termed ceragenins or CSAs have been developed, primarily by Dr. Paul B. Savage. These compounds reproduce the amphiphilic morphology in antimicrobial peptides, represented above by CSA-13, and display potent, as well as diverse, biological activities (including, but not limited to anti-bacterial, anti-cancer, anti-inflammatory, promoting bone growth, promoting wound healing, etc.). Lead ceragenins can be produced at a large scale, and because they are not peptide based, they are not substrates for proteases. Consequently, the ceragenins represented an attractive compound class for producing pharmaceutically relevant treatments.

SUMMARY

[0007] Embodiments described herein relate to sulfonic acid addition salt of a CSA. In preferred embodiments, the sulfonic acid addition salt is a di-addition salt of a CSA and 1,5-naphthalenedisulfonic acid ("NDSA").

[0008] In some embodiments, the sulfonic acid addition salt is a solid. In some embodiments, the solid is a flowable solid. In some embodiments, the sulfonic acid addition salt is crystalline. In some embodiments, the sulfonic acid addition salt is storage stable. In some embodiments, the sulfonic acid addition salt is micronized.

[0009] Preferred embodiments provide a formulation comprising a di-addition salt of a CSA and NDSA and a pharmaceutically acceptable excipient.

[0010] Some embodiments provide a process for preparing a CSA salt, comprising diluting the free base of a CSA with a solvent; adding at least one equivalent of an acid to the diluted CSA in solvent to afford a reaction mixture; precipitating or temperature cycling the reaction mixture; and isolating a CSA salt.

[0011] In some embodiments, the temperature cycling is conducted for at least about 48 hours. In some embodiments, the process further comprises utilizing an anti-solvent or evaporation of solvent when isolating the CSA salt.

[0012] In some embodiments, the CSA salt is a solid. In some embodiments, the CSA salt is crystalline. In some embodiments, the CSA salt is amorphous. In some embodiments, the CSA salt is storage stable. In some embodiments, the CSA salt is flowable. In some embodiments, the CSA salt is micronized.

[0013] Advantages of the CSA compounds disclosed herein include, but are not limited to, comparable and/or improved antimicrobial activity, stability, and/or pharmaceutical administerability compared to existing CSA compounds and/or simplified synthesis of final CSA compounds and/or intermediate CSA compounds compared to existing synthetic routes.

[0014] Additional features and advantages will be set forth in part in the description that follows, and in part will be obvious from the description, or may be learned by practice of the embodiments disclosed herein. It is to be understood that both the foregoing brief summary and the following detailed description are exemplary and explanatory only and are not restrictive of the embodiments disclosed herein or as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] To further clarify the above and other advantages and features of the present invention, a more particular description of the invention will be rendered by reference to specific embodiments thereof which are illustrated in the appended drawings. It is appreciated that these drawings depict only illustrated embodiments of the invention and are therefore not to be considered limiting of its scope. The invention will be described and explained with additional specificity and detail through the use of the accompanying drawings in which:

[0016] FIGS. 1-3 illustrate x-ray powder diffraction (XRPD) spectrum of CSA mono-sulfate salt compounds according to the present disclosure;

[0017] FIG. 4 illustrates an x-ray powder diffraction (XRPD) spectrum of CSA di-fumarate salt compounds according to the present disclosure;

[0018] FIGS. 5 and 6 illustrate x-ray powder diffraction (XRPD) spectrum of CSA di-NDSA salt compounds according to the present disclosure;

[0019] FIG. 7 illustrates a dynamic vapor sorption (DVS) isotherm plot of a CSA di-NDSA salt compound of the present disclosure;

[0020] FIG. 8 illustrates an XRPD spectrum of a CSA di-NDSA salt compound after being

subjected to a DVS analysis;

[0021] FIG. **9** illustrates an overlay of XRPD spectrums of a CSA di-NDSA compound salt composition embodiment showing results before and after DVS analysis of the salt composition; and

[0022] FIG. **10** contains graphics comparing the amount of biofilm and reduction in biofilm between endotracheal tube (ETT) segments that were either uncoated (black bar) or coated with polyurethane containing the di-NDSA salt of CSA-131 (grey bar).

DETAILED DESCRIPTION

[0023] The embodiments disclosed herein will now be described by reference to some more detailed embodiments, with occasional reference to any applicable accompanying drawings. These embodiments may, however, be embodied in different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the embodiments to those skilled in the art.

Definitions

[0024] Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which these embodiments belong. The terminology used in the description herein is for describing particular embodiments only and is not intended to be limiting of the embodiments. As used in the specification and the appended claims, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. All publications, patent applications, patents, and other references mentioned herein are incorporated by reference in their entirety.

[0025] Terms and phrases used in this application, and variations thereof, especially in the appended claims, unless otherwise expressly stated, should be construed as open ended as opposed to limiting. As examples of the foregoing, the term “including” should be read to mean “including, without limitation,” “including but not limited to,” or the like; the term “comprising” as used herein is synonymous with “including,” “containing,” or “characterized by,” and is inclusive or open-ended and does not exclude additional, unrecited elements or method steps; the term “having” should be interpreted as “having at least”; the term “includes” should be interpreted as “includes but is not limited to”; the term “example” is used to provide exemplary instances of the item in discussion, not an exhaustive or limiting list thereof; and use of terms like “preferably,” “preferred,” “desired,” or “desirable,” and words of similar meaning should not be understood as implying that certain features are critical, essential, or even important to the structure or function of the invention, but instead as merely intended to highlight alternative or additional features that may or may not be utilized in a particular embodiment. In addition, the term “comprising” is to be interpreted synonymously with the phrases “having at least” or “including at least”. When used in the context of a process, the term “comprising” means that the process includes at least the recited steps, but may include additional steps. When used in the context of a compound, composition or device, the term “comprising” means that the compound, composition or device includes at least the recited features or components, but may also include additional features or components. Likewise, a group of items linked with the conjunction “and” should not be read as requiring that each and every one of those items be present in the grouping, but rather should be read as “and/or” unless expressly stated otherwise. Similarly, a group of items linked with the conjunction “or” should not be read as requiring mutual exclusivity among that group, but rather should be read as “and/or” unless expressly stated otherwise.

[0026] It is understood that, in any compound described herein having one or more chiral centers, if an absolute stereochemistry is not expressly indicated, then each center may independently be of R-configuration or S-configuration or a mixture thereof. Thus, the compounds provided herein may be enantiomerically pure, enantiomerically enriched, racemic mixture, diastereomerically pure, diastereomerically enriched, or a stereoisomeric mixture. In addition it is understood that, in any

compound described herein having one or more double bond(s) generating geometrical isomers that can be defined as E or Z, each double bond may independently be E or Z a mixture thereof.

[0027] Likewise, it is understood that, in any compound described, all tautomeric forms are also intended to be included.

[0028] It is to be understood that where compounds disclosed herein have unfilled valencies, then the valencies are to be filled with hydrogens or isotopes thereof, e.g., hydrogen-1 (protium) and hydrogen-2 (deuterium).

[0029] It is understood that the compounds described herein can be labeled isotopically.

Substitution with isotopes such as deuterium may afford certain therapeutic advantages resulting from greater metabolic stability, such as, for example, increased in vivo half-life or reduced dosage requirements. Each chemical element as represented in a compound structure may include any isotope of said element. For example, in a compound structure a hydrogen atom may be explicitly disclosed or understood to be present in the compound. At any position of the compound that a hydrogen atom may be present, the hydrogen atom can be any isotope of hydrogen, including but not limited to hydrogen-1 (protium) and hydrogen-2 (deuterium). Thus, reference herein to a compound encompasses all potential isotopic forms unless the context clearly dictates otherwise.

[0030] Unless otherwise indicated, all numbers expressing quantities of ingredients, reaction conditions, and so forth used in the specification and claims are to be understood as being modified in all instances by the term “about.” Accordingly, unless indicated to the contrary, the numerical parameters set forth in the specification and attached claims are approximations that may vary depending upon the desired properties sought to be obtained by the present embodiments. At the very least, and not as an attempt to limit the application of the doctrine of equivalents to the scope of the claims, each numerical parameter should be construed in light of the number of significant digits and ordinary rounding approaches.

[0031] Notwithstanding that the numerical ranges and parameters setting forth the broad scope of the embodiments are approximations, the numerical values set forth in the specific examples are reported as precisely as possible. Any numerical value, however, inherently contains certain errors necessarily resulting from the standard deviation found in their respective testing measurements. Every numerical range given throughout this specification and claims will include every narrower numerical range that falls within such broader numerical range, as if such narrower numerical ranges were all expressly written herein. Where a range of values is provided, it is understood that the upper and lower limit, and each intervening value between the upper and lower limit of the range is encompassed within the embodiments.

[0032] As used herein, any “R” group(s) such as, without limitation, R.sub.1, R.sub.2, R.sub.3, R.sub.4, R.sub.5, R.sub.6, R.sub.7, R.sub.8, R.sub.9, R.sub.10, R.sub.11, R.sub.12, R.sub.13, R.sub.14, R.sub.15, R.sub.16, R.sub.17, and R.sub.18 represent substituents that can be attached to the indicated atom. Unless otherwise specified, an R group may be substituted or unsubstituted.

[0033] A “ring” as used herein can be heterocyclic or carbocyclic. The term “saturated” used herein refers to a ring having each atom in the ring either hydrogenated or substituted such that the valency of each atom is filled. The term “unsaturated” used herein refers to a ring where the valency of each atom of the ring may not be filled with hydrogen or other substituents. For example, adjacent carbon atoms in the fused ring can be doubly bound to each other. Unsaturation can also include deleting at least one of the following pairs and completing the valency of the ring carbon atoms at these deleted positions with a double bond, such as R.sub.5 and R.sub.9; R.sub.8 and R.sub.10; and R.sub.13 and R.sub.14.

[0034] Whenever a group is described as being “substituted” that group may be substituted with one, two, three or more of the indicated substituents, which may be the same or different, each replacing a hydrogen atom. If no substituents are indicated, it is meant that the indicated “substituted” group may be substituted with one or more group(s) individually and independently selected from alkyl, alkenyl, alkynyl, cycloalkyl, cycloalkenyl, cycloalkynyl, acylalkyl,

alkoxyalkyl, aminoalkyl, amino acid, aryl, heteroaryl, heteroalicycyl, aralkyl, heteroaralkyl, (heteroalicycyl)alkyl, hydroxy, protected hydroxyl, alkoxy, aryloxy, acyl, mercapto, alkylthio, arylthio, cyano, halogen (e.g., F, Cl, Br, and I), thiocarbonyl, O-carbamyl, N-carbamyl, O-thiocarbamyl, N-thiocarbamyl, C-amido, N-amido, S-sulfonamido, N-sulfonamido, C-carboxy, protected C-carboxy, O-carboxy, isocyanato, thiocyanato, isothiocyanato, nitro, oxo, silyl, sulfenyl, sulfinyl, sulfonyl, haloalkyl, haloalkoxy, trihalomethanesulfonyl, trihalomethanesulfonamido, an amino, a mono-substituted amino group and a di-substituted amino group, R.sub.aO(CH.sub.2).sub.mO—, R.sub.b(CH.sub.2).sub.nO—, R.sub.cC(O)O(CH.sub.2).sub.pO—, and protected derivatives thereof. The substituent may be attached to the group at more than one attachment point. For example, an aryl group may be substituted with a heteroaryl group at two attachment points to form a fused multicyclic aromatic ring system. Biphenyl and naphthalene are two examples of an aryl group that is substituted with a second aryl group. A group that is not specifically labeled as substituted or unsubstituted may be considered to be either substituted or unsubstituted.

[0035] As used herein, “C.sub.a” or “C.sub.a to C.sub.b” in which “a” and “b” are integers refer to the number of carbon atoms in an alkyl, alkenyl or alkynyl group, or the number of carbon atoms in the ring of a cycloalkyl, cycloalkenyl, cycloalkynyl, aryl, heteroaryl or heteroalicycyl group. That is, the alkyl, alkenyl, alkynyl, ring of the cycloalkyl, ring of the cycloalkenyl, ring of the cycloalkynyl, ring of the aryl, ring of the heteroaryl or ring of the heteroalicycyl can contain from “a” to “b”, inclusive, carbon atoms. Thus, for example, a “C.sub.1 to C.sub.4 alkyl” group refers to all alkyl groups having from 1 to 4 carbons, that is, CH.sub.3—, CH.sub.3CH.sub.2—, CH.sub.3CH.sub.2CH.sub.2—, (CH.sub.3).sub.2CH—, CH.sub.3CH.sub.2CH.sub.2CH.sub.2—, CH.sub.3CH.sub.2CH(CH.sub.3)— and (CH.sub.3).sub.3C—. If no “a” and “b” are designated with regard to an alkyl, alkenyl, alkynyl, cycloalkyl cycloalkenyl, cycloalkynyl, aryl, heteroaryl or heteroalicycyl group, the broadest range described in these definitions is to be assumed.

[0036] As used herein, “alkyl” refers to a straight or branched hydrocarbon chain that comprises a fully saturated (no double or triple bonds) hydrocarbon group. The alkyl group may have 1 to 25 carbon atoms (whenever it appears herein, a numerical range such as “1 to 25” refers to each integer in the given range; e.g., “1 to 25 carbon atoms” means that the alkyl group may consist of 1 carbon atom, 2 carbon atoms, 3 carbon atoms, etc., up to and including 25 carbon atoms, although the present definition also covers the occurrence of the term “alkyl” where no numerical range is designated). The alkyl group may also be a medium size alkyl having 1 to 15 carbon atoms. The alkyl group could also be a lower alkyl having 1 to 6 carbon atoms. The alkyl group of the compounds may be designated as “C.sub.4” or “C.sub.1-C.sub.4 alkyl” or similar designations. By way of example only, “C.sub.1-C.sub.4 alkyl” indicates that there are one to four carbon atoms in the alkyl chain, i.e., the alkyl chain is selected from methyl, ethyl, propyl, iso-propyl, n-butyl, iso-butyl, sec-butyl, and t-butyl. Typical alkyl groups include, but are in no way limited to, methyl, ethyl, propyl, isopropyl, butyl, isobutyl, tertiary butyl, pentyl and hexyl. The alkyl group may be substituted or unsubstituted.

[0037] As used herein, “alkenyl” refers to an alkyl group that contains in the straight or branched hydrocarbon chain one or more double bonds. The alkenyl group may have 2 to 25 carbon atoms (whenever it appears herein, a numerical range such as “2 to 25” refers to each integer in the given range; e.g., “2 to 25 carbon atoms” means that the alkenyl group may consist of 2 carbon atom, 3 carbon atoms, 4 carbon atoms, etc., up to and including 25 carbon atoms, although the present definition also covers the occurrence of the term “alkenyl” where no numerical range is designated). The alkenyl group may also be a medium size alkenyl having 2 to 15 carbon atoms. The alkenyl group could also be a lower alkenyl having 1 to 6 carbon atoms. The alkenyl group of the compounds may be designated as “C.sub.4” or “C.sub.2-C.sub.4 alkyl” or similar designations. An alkenyl group may be unsubstituted or substituted.

[0038] As used herein, “alkynyl” refers to an alkyl group that contains in the straight or branched

hydrocarbon chain one or more triple bonds. The alkynyl group may have 2 to 25 carbon atoms (whenever it appears herein, a numerical range such as “2 to 25” refers to each integer in the given range; e.g., “2 to 25 carbon atoms” means that the alkynyl group may consist of 2 carbon atom, 3 carbon atoms, 4 carbon atoms, etc., up to and including 25 carbon atoms, although the present definition also covers the occurrence of the term “alkynyl” where no numerical range is designated). The alkynyl group may also be a medium size alkynyl having 2 to 15 carbon atoms. The alkynyl group could also be a lower alkynyl having 2 to 6 carbon atoms. The alkynyl group of the compounds may be designated as “C.sub.4” or “C.sub.2-C.sub.4 alkyl” or similar designations. An alkynyl group may be unsubstituted or substituted.

[0039] As used herein, “aryl” refers to a carbocyclic (all carbon) monocyclic or multicyclic aromatic ring system (including fused ring systems where two carbocyclic rings share a chemical bond) that has a fully delocalized pi-electron system throughout all the rings. The number of carbon atoms in an aryl group can vary. For example, the aryl group can be a C.sub.6-C.sub.14 aryl group, a C.sub.6-C.sub.10 aryl group, or a C.sub.6 aryl group (although the definition of C.sub.6-C.sub.10 aryl covers the occurrence of “aryl” when no numerical range is designated). Examples of aryl groups include, but are not limited to, benzene, naphthalene and azulene. An aryl group may be substituted or unsubstituted.

[0040] As used herein, “aralkyl” and “aryl(alkyl)” refer to an aryl group connected, as a substituent, via a lower alkylene group. The aralkyl group may have 6 to 20 carbon atoms (whenever it appears herein, a numerical range such as “6 to 20” refers to each integer in the given range; e.g., “6 to 20 carbon atoms” means that the aralkyl group may consist of 6 carbon atom, 7 carbon atoms, 8 carbon atoms, etc., up to and including 20 carbon atoms, although the present definition also covers the occurrence of the term “aralkyl” where no numerical range is designated). The lower alkylene and aryl group of an aralkyl may be substituted or unsubstituted. Examples include but are not limited to benzyl, 2-phenylalkyl, 3-phenylalkyl, and naphthylalkyl.

[0041] “Lower alkylene groups” refer to a C.sub.1-C.sub.25 straight-chained alkyl tethering groups, such as —CH.sub.2-tethering groups, forming bonds to connect molecular fragments via their terminal carbon atoms. Examples include but are not limited to methylene (—CH.sub.2—), ethylene (—CH.sub.2CH.sub.2—), propylene (—CH.sub.2CH.sub.2CH.sub.2—), and butylene (—CH.sub.2CH.sub.2CH.sub.2CH.sub.2—). A lower alkylene group can be substituted by replacing one or more hydrogen of the lower alkylene group with a substituent(s) listed under the definition of “substituted.”

[0042] As used herein, “cycloalkyl” refers to a completely saturated (no double or triple bonds) mono- or multi-cyclic hydrocarbon ring system. When composed of two or more rings, the rings may be joined together in a fused fashion. Cycloalkyl groups can contain 3 to 10 atoms in the ring(s) or 3 to 8 atoms in the ring(s). A cycloalkyl group may be unsubstituted or substituted. Typical cycloalkyl groups include, but are in no way limited to, cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, cycloheptyl and cyclooctyl.

[0043] As used herein, “cycloalkenyl” refers to a mono- or multi-cyclic hydrocarbon ring system that contains one or more double bonds in at least one ring; although, if there is more than one, the double bonds cannot form a fully delocalized pi-electron system throughout all the rings (otherwise the group would be “aryl,” as defined herein). When composed of two or more rings, the rings may be connected together in a fused fashion. A cycloalkenyl group may be unsubstituted or substituted.

[0044] As used herein, “cycloalkynyl” refers to a mono- or multi-cyclic hydrocarbon ring system that contains one or more triple bonds in at least one ring. If there is more than one triple bond, the triple bonds cannot form a fully delocalized pi-electron system throughout all the rings. When composed of two or more rings, the rings may be joined together in a fused fashion. A cycloalkynyl group may be unsubstituted or substituted.

[0045] As used herein, “alkoxy” or “alkyloxy” refers to the formula —OR wherein R is an alkyl, an alkenyl, an alkynyl, a cycloalkyl, a cycloalkenyl or a cycloalkynyl as defined above. A non-limiting

list of alkoxys are methoxy, ethoxy, n-propoxy, 1-methylethoxy (isopropoxy), n-butoxy, iso-butoxy, sec-butoxy and tert-butoxy. An alkoxy may be substituted or unsubstituted.

[0046] As used herein, “acyl” refers to a hydrogen, alkyl, alkenyl, alkynyl, aryl, or heteroaryl connected, as substituents, via a carbonyl group. Examples include formyl, acetyl, propanoyl, benzoyl, and acryl. An acyl may be substituted or unsubstituted.

[0047] As used herein, “alkoxyalkyl” or “alkyloxyalkyl” refers to an alkoxy group connected, as a substituent, via a lower alkylene group. Examples include alkyl-O-alkyl- and alkoxy-alkyl- with the terms alkyl and alkoxy defined herein.

[0048] As used herein, “hydroxyalkyl” refers to an alkyl group in which one or more of the hydrogen atoms are replaced by a hydroxy group. Exemplary hydroxyalkyl groups include but are not limited to, 2-hydroxyethyl, 3-hydroxypropyl, 2-hydroxypropyl, and 2,2-dihydroxyethyl. A hydroxyalkyl may be substituted or unsubstituted.

[0049] As used herein, “haloalkyl” refers to an alkyl group in which one or more of the hydrogen atoms are replaced by a halogen (e.g., mono-haloalkyl, di-haloalkyl and tri-haloalkyl). Such groups include but are not limited to, chloromethyl, fluoromethyl, difluoromethyl, trifluoromethyl and 1-chloro-2-fluoromethyl, 2-fluoroisobutyl. A haloalkyl may be substituted or unsubstituted.

[0050] The term “amino” as used herein refers to a —NH₂ group.

[0051] As used herein, the term “hydroxy” refers to a —OH group.

[0052] A “cyano” group refers to a “—CN” group.

[0053] A “carbonyl” or an “oxo” group refers to a C=O group.

[0054] The term “azido” as used herein refers to a —N₃ group.

[0055] As used herein, “aminoalkyl” refers to an amino group connected, as a substituent, via a lower alkylene group. Examples include H₂N-alkyl- with the term alkyl defined herein.

[0056] As used herein, “alkylcarboxyalkyl” refers to an alkyl group connected, as a substituent, to a carboxy group that is connected, as a substituent, to an alkyl group. Examples include alkyl-C(=O)O-alkyl- and alkyl-O—C(=O)-alkyl- with the term alkyl as defined herein.

[0057] As used herein, “alkylaminoalkyl” refers to an alkyl group connected, as a substituent, to an amino group that is connected, as a substituent, to an alkyl group. Examples include alkyl-NH-alkyl-, with the term alkyl as defined herein.

[0058] As used herein, “dialkylaminoalkyl” or “di(alkyl)aminoalkyl” refers to two alkyl groups connected, each as a substituent, to an amino group that is connected, as a substituent, to an alkyl group. Examples include

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with the term alkyl as defined herein.

[0059] As used herein, “alkylaminoalkylamino” refers to an alkyl group connected, as a substituent, to an amino group that is connected, as a substituent, to an alkyl group that is connected, as a substituent, to an amino group. Examples include alkyl-NH-alkyl-NH—, with the term alkyl as defined herein.

[0060] As used herein, “alkylaminoalkylaminoalkylamino” refers to an alkyl group connected, as a substituent, to an amino group that is connected, as a substituent, to an alkyl group that is connected, as a substituent, to an amino group that is connected, as a substituent, to an alkyl group. Examples include alkyl-NH-alkyl-NH-alkyl-, with the term alkyl as defined herein.

[0061] As used herein, “arylaminoalkyl” refers to an aryl group connected, as a substituent, to an amino group that is connected, as a substituent, to an alkyl group. Examples include aryl-NH-alkyl-, with the terms aryl and alkyl as defined herein.

[0062] As used herein, “aminoalkyloxy” refers to an amino group connected, as a substituent, to an alkyloxy group. Examples include H₂N-alkyl-O— and H₂N-alkoxy- with the terms alkyl and alkoxy as defined herein.

[0063] As used herein, “aminoalkyloxyalkyl” refers to an amino group connected, as a substituent, to an alkyloxy group connected, as a substituent, to an alkyl group. Examples include H₂N-

alkyl-O-alkyl- and H.sub.2N-alkoxy-alkyl- with the terms alkyl and alkoxy as defined herein.

[0064] As used herein, “aminoalkylcarboxy” refers to an amino group connected, as a substituent, to an alkyl group connected, as a substituent, to a carboxy group. Examples include H.sub.2N-alkyl-C(=O)O— and H.sub.2N-alkyl-O—C(=O)— with the term alkyl as defined herein.

[0065] As used herein, “aminoalkylaminocarbonyl” refers to an amino group connected, as a substituent, to an alkyl group connected, as a substituent, to an amino group connected, as a substituent, to a carbonyl group. Examples include H.sub.2N-alkyl-NH—C(=O)— with the term alkyl as defined herein.

[0066] As used herein, “aminoalkylcarboxamido” refers to an amino group connected, as a substituent, to an alkyl group connected, as a substituent, to a carbonyl group connected, as a substituent to an amino group. Examples include H.sub.2N-alkyl-C(=O)—NH— with the term alkyl as defined herein.

[0067] As used herein, “azidoalkyloxy” refers to an azido group connected as a substituent, to an alkyloxy group. Examples include N.sub.3-alkyl-O— and N.sub.3-alkoxy- with the terms alkyl and alkoxy as defined herein.

[0068] As used herein, “cyanoalkyloxy” refers to a cyano group connected as a substituent, to an alkyloxy group. Examples include NC-alkyl-O— and NC-alkoxy- with the terms alkyl and alkoxy as defined herein.

[0069] A “sulfenyl” group refers to an “—SR” group in which R can be hydrogen, alkyl, alkenyl, alkynyl, cycloalkyl, cycloalkenyl, cycloalkynyl, aryl, heteroaryl, heteroalicycyl, aralkyl, or (heteroalicycyl)alkyl. A sulfenyl may be substituted or unsubstituted.

[0070] A “sulfinyl” group refers to an “—S(=O)—R” group in which R can be the same as defined with respect to sulfenyl. A sulfinyl may be substituted or unsubstituted.

[0071] A “sulfonyl” group refers to an “SO.sub.2R” group in which R can be the same as defined with respect to sulfenyl. A sulfonyl may be substituted or unsubstituted.

[0072] An “O-carboxy” group refers to a “RC(=O)O—” group in which R can be hydrogen, alkyl, alkenyl, alkynyl, cycloalkyl, cycloalkenyl, cycloalkynyl, aryl, heteroaryl, heteroalicycyl, aralkyl, or (heteroalicycyl)alkyl, as defined herein. An O-carboxy may be substituted or unsubstituted.

[0073] The terms “ester” and “C-carboxy” refer to a “—C(=O)OR” group in which R can be the same as defined with respect to O-carboxy. An ester and C-carboxy may be substituted or unsubstituted.

[0074] A “thiocarbonyl” group refers to a “—C(=S)R” group in which R can be the same as defined with respect to O-carboxy. A thiocarbonyl may be substituted or unsubstituted.

[0075] A “trihalomethanesulfonyl” group refers to an “X.sub.3CSO.sub.2—” group wherein X is a halogen.

[0076] An “S-sulfonamido” group refers to a “—SO.sub.2N(RARB)” group in which RA and RB can be independently hydrogen, alkyl, alkenyl, alkynyl, cycloalkyl, cycloalkenyl, cycloalkynyl, aryl, heteroaryl, heteroalicycyl, aralkyl, or (heteroalicycyl)alkyl. An S-sulfonamido may be substituted or unsubstituted.

[0077] An “N-sulfonamido” group refers to a “RSO.sub.2N(RA)-” group in which R and RA can be independently hydrogen, alkyl, alkenyl, alkynyl, cycloalkyl, cycloalkenyl, cycloalkynyl, aryl, heteroaryl, heteroalicycyl, aralkyl, or (heteroalicycyl)alkyl. An N-sulfonamido may be substituted or unsubstituted.

[0078] An “O-carbamyl” group refers to a “—OC(=O)N(RARB)” group in which RA and RB can be independently hydrogen, alkyl, alkenyl, alkynyl, cycloalkyl, cycloalkenyl, cycloalkynyl, aryl, heteroaryl, heteroalicycyl, aralkyl, or (heteroalicycyl)alkyl. An O-carbamyl may be substituted or unsubstituted.

[0079] An “N-carbamyl” group refers to an “ROC(=O)N(RA)-” group in which R and RA can be independently hydrogen, alkyl, alkenyl, alkynyl, cycloalkyl, cycloalkenyl, cycloalkynyl, aryl, heteroaryl, heteroalicycyl, aralkyl, or (heteroalicycyl)alkyl. An N-carbamyl may be substituted or

unsubstituted.

[0080] An “O-thiocarbamyl” group refers to a “—OC(=S)—N(RARB)” group in which RA and RB can be independently hydrogen, alkyl, alkenyl, alkynyl, cycloalkyl, cycloalkenyl, cycloalkynyl, aryl, heteroaryl, heteroalicycyl, aralkyl, or (heteroalicycyl)alkyl. An O—thiocarbamyl may be substituted or unsubstituted.

[0081] An “N-thiocarbamyl” group refers to an “ROC(=S)N(RA)-” group in which R and RA can be independently hydrogen, alkyl, alkenyl, alkynyl, cycloalkyl, cycloalkenyl, cycloalkynyl, aryl, heteroaryl, heteroalicycyl, aralkyl, or (heteroalicycyl)alkyl. An N-thiocarbamyl may be substituted or unsubstituted.

[0082] A “C-amido” group refers to a “—C(=O)N(RARB)” group in which RA and RB can be independently hydrogen, alkyl, alkenyl, alkynyl, cycloalkyl, cycloalkenyl, cycloalkynyl, aryl, heteroaryl, heteroalicycyl, aralkyl, or (heteroalicycyl)alkyl. A C-amido may be substituted or unsubstituted.

[0083] An “N-amido” group refers to a “RC(=O)N(RA)-” group in which R and RA can be independently hydrogen, alkyl, alkenyl, alkynyl, cycloalkyl, cycloalkenyl, cycloalkynyl, aryl, heteroaryl, heteroalicycyl, aralkyl, or (heteroalicycyl)alkyl. An N-amido may be substituted or unsubstituted.

[0084] As used herein, “guanidinoalkyloxy” refers to a guanidiny group connected, as a substituent, to an alkyloxy group. Examples include

##STR00002##

with the terms alkyl and alkoxy as defined herein.

[0085] As used herein, “guanidinoalkylcarboxy” refers to a guanidiny group connected, as a substituent, to an alkyl group connected, as a substituent, to a carboxy group. Examples include

##STR00003##

with the term alkyl as defined herein.

[0086] As used herein, “quaternary ammonium alkylcarboxy” refers to a quaternized amino group connected, as a substituent, to an alkyl group connected, as a substituent, to a carboxy group.

Examples include

##STR00004##

with the term alkyl as defined herein.

[0087] The term “halogen atom” or “halogen” as used herein, means any one of the radio-stable atoms of column 7 of the Periodic Table of the Elements, such as, fluorine, chlorine, bromine and iodine.

[0088] Where the numbers of substituents is not specified (e.g. haloalkyl), there may be one or more substituents present. For example “haloalkyl” may include one or more of the same or different halogens.

[0089] As used herein, the term “amino acid” refers to any amino acid (both standard and non-standard amino acids), including, but not limited to, α -amino acids, β -amino acids, γ -amino acids and δ -amino acids. Examples of suitable amino acids include, but are not limited to, alanine, asparagine, aspartate, cysteine, glutamate, glutamine, glycine, proline, serine, tyrosine, arginine, histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan and valine. Additional examples of suitable amino acids include, but are not limited to, ornithine, hypusine, 2-aminoisobutyric acid, dehydroalanine, gamma-aminobutyric acid, citrulline, beta-alanine, alpha-ethyl-glycine, alpha-propyl-glycine and norleucine.

[0090] A linking group is a divalent moiety used to link one steroid to another steroid. In some embodiments, the linking group is used to link a first CSA with a second CSA (which may be the same or different). An example of a linking group is (C.sub.1-C.sub.10) alkyloxy-(C.sub.1-C.sub.10) alkyl.

[0091] The terms “P.G.” or “protecting group” or “protecting groups” as used herein refer to any atom or group of atoms that is added to a molecule in order to prevent existing groups in the

molecule from undergoing unwanted chemical reactions. Examples of protecting group moieties are described in T. W. Greene and P. G. M. Wuts, *Protective Groups in Organic Synthesis*, 3. Ed. John Wiley & Sons, 1999, and in J. F. W. McOmie, *Protective Groups in Organic Chemistry* Plenum Press, 1973, both of which are hereby incorporated by reference for the limited purpose of disclosing suitable protecting groups. The protecting group moiety may be chosen in such a way, that they are stable to certain reaction conditions and readily removed at a convenient stage using methodology known from the art. A non-limiting list of protecting groups include benzyl; substituted benzyl; alkylcarbonyls and alkoxycarbonyls (e.g., t-butoxycarbonyl (BOC), acetyl, or isobutyryl); arylalkylcarbonyls and arylalkoxycarbonyls (e.g., benzyloxycarbonyl); substituted methyl ether (e.g. methoxymethyl ether); substituted ethyl ether; a substituted benzyl ether; tetrahydropyranyl ether; silyls (e.g., trimethylsilyl, triethylsilyl, triisopropylsilyl, t-butyltrimethylsilyl, tri-iso-propylsilyloxymethyl, [2-(trimethylsilyl)ethoxy]methyl or t-butyltriphenylsilyl); esters (e.g. benzoate ester); carbonates (e.g. methoxymethylcarbonate); sulfonates (e.g. tosylate or mesylate); acyclic ketal (e.g. dimethyl acetal); cyclic ketals (e.g., 1,3-dioxane, 1,3-dioxolanes, and those described herein); acyclic acetal; cyclic acetal (e.g., those described herein); acyclic hemiacetal; cyclic hemiacetal; cyclic dithioketals (e.g., 1,3-dithiane or 1,3-dithiolane); orthoesters (e.g., those described herein) and triarylmethyl groups (e.g., trityl; monomethoxytrityl (MMTr); 4,4'-dimethoxytrityl (DMTr); 4,4',4''-trimethoxytrityl (TMTr); and those described herein). Amino-protecting groups are known to those skilled in the art. In general, the species of protecting group is not critical, provided that it is stable to the conditions of any subsequent reaction(s) on other positions of the compound and can be removed at the appropriate point without adversely affecting the remainder of the molecule. In addition, a protecting group may be substituted for another after substantive synthetic transformations are complete. Clearly, where a compound differs from a compound disclosed herein only in that one or more protecting groups of the disclosed compound has been substituted with a different protecting group, that compound is within the disclosure.

CSA Compounds

[0092] Cationic steroidal anti-microbial (CSA) compounds, sometimes referred to as “CSA compounds” or “ceragenin” compounds, are synthetically produced, small molecule chemical compounds that include a sterol backbone having various charged groups (e.g., amine and cationic groups) attached to the backbone. The sterol backbone can be used to orient amine or guanidine groups on a face or plane of the sterol backbone. CSAs are cationic and amphiphilic, based upon the functional groups attached to the backbone. They are facially amphiphilic with a hydrophobic face and a polycationic face.

[0093] Without wishing to be bound to theory, the CSA molecules described herein act as anti-microbial agents (e.g., anti-bacterial, anti-fungal, and anti-viral). It is believed, for example, that anti-microbial CSA molecules may act as an anti-microbial by binding to the cellular membrane of bacteria and other microbes and modifying the cell membrane, e.g., such as by forming a pore that allows the leakage of ions and cytoplasmic materials critical to the microbe's survival, and leading to the death of the affected microbe. In addition, anti-microbial CSA molecules may also act to sensitize bacteria to other antibiotics. For example, at concentrations of anti-microbial CSA molecules below the corresponding minimum bacteriostatic concentration (MIC), the CSA compound may cause bacteria to become more susceptible to other antibiotics by disrupting the cell membrane, such as by increasing membrane permeability. It is postulated that charged cationic groups may be responsible for disrupting the bacterial cellular membrane and imparting anti-microbial properties. CSA molecules may have similar membrane- or outer coating-disrupting effects on fungi and viruses.

[0094] Compounds useful in accordance with this disclosure are described herein, both generically and with particularity, and in U.S. Pat. Nos. 6,350,738, 6,486,148, 6,767,904, 7,598,234, 7,754,705, U.S. Application Nos. 61/786,301, 13/288,892, 61/642431, 13/554,930, 61/572,714, 13/594,608,

61/576,903, 13/594,612, 13/288,902, 61/605,639, 13/783,131, 61/605,642, 13/783,007, 61/132,361, 13/000,010, 61/534,185, 13/615,244, 61/534,194, 13/615,324, 61/534,205, 61/637,402, 13/841,549, 61/715,277, PCT/US13/37615, 61/749,800, 61/794,721, and 61/814,816, which are incorporated herein by reference. The skilled artisan will recognize the compounds within the generic formula set forth herein and understand their preparation in view of the references cited herein and the Examples.

[0095] In some embodiments, CSA compounds as disclosed herein can be a compound of Formula (I), Formula (II), or salt thereof, having a steroidal backbone:

##STR00005##

[0096] CSA compounds of Formula (I), Formula (II), and salts thereof can be characterized wherein: [0097] rings A, B, C, and D are independently saturated, or are fully or partially unsaturated, provided that at least two of rings A, B, C, and D are saturated; [0098] m, n, p, and q are independently 0 or 1; [0099] R.sub.1 through R.sub.4, R.sub.6, R.sub.7, R.sub.11, R.sub.12, R.sub.15, R.sub.16, and R.sub.18 are independently selected from the group consisting of hydrogen, hydroxyl, alkyl, hydroxyalkyl, alkyloxyalkyl, alkylcarboxyalkyl, alkylaminoalkyl, alkylaminoalkylamino, alkylaminoalkylaminoalkylamino, aminoalkyl, aryl, arylaminoalkyl, haloalkyl, alkenyl, alkynyl, oxo, a linking group attached to a second steroid, aminoalkyloxy, aminoalkyloxyalkyl, aminoalkylcarboxy, aminoalkylaminocarbonyl, aminoalkylcarboxamido, di(alkyl)aminoalkyl, H.sub.2N—HC(Q.sub.5)-C(O)—O—, H.sub.2N—HC(Q.sub.5)-C(O)—N(H)—, azidoalkyloxy, cyanoalkyloxy, P.G.-HN—HC(Q.sub.5)-C(O)—O—, guanidinoalkyloxy, quaternary ammonium alkylcarboxy, and guanidinoalkyl carboxy, where Q.sub.5 is a side chain of any amino acid (including a side chain of glycine, i.e., H), and P.G. is an amino protecting group; and [0100] R.sub.5, R.sub.8, R.sub.9, R.sub.10, R.sub.13, R.sub.14 and R.sub.17 are independently deleted when one of rings A, B, C, or D is unsaturated so as to complete the valency of the carbon atom at that site, or R.sub.5, R.sub.8, R.sub.9, R.sub.10, R.sub.13, and R.sub.14 are independently selected from the group consisting of hydrogen, hydroxyl, alkyl, hydroxyalkyl, alkyloxyalkyl, aminoalkyl, aryl, haloalkyl, alkenyl, alkynyl, oxo, a linking group attached to a second steroid, aminoalkyloxy, aminoalkylcarboxy, aminoalkylaminocarbonyl, di(alkyl)aminoalkyl, H.sub.2N—HC(Q.sub.5)-C(O)—O—, H.sub.2N—HC(Q.sub.5)-C(O)—N(H)—, azidoalkyloxy, cyanoalkyloxy, P.G.-HN—HC(Q.sub.5)-C(O)—O—, guanidinoalkyloxy, and guanidinoalkyl-carboxy, where Q.sub.5 is a side chain of any amino acid, P.G. is an amino protecting group.

[0101] In some embodiments, at least one, and sometimes two or three of R.sub.1-4, R.sub.6, R.sub.7, R.sub.11, R.sub.12, R.sub.15, R.sub.16, R.sub.17, and R.sub.18 are independently selected from the group consisting of aminoalkyl, aminoalkyloxy, alkylcarboxyalkyl, alkylaminoalkylamino, alkylaminoalkyl-aminoalkylamino, aminoalkylcarboxy, arylaminoalkyl, aminoalkyloxyaminoalkylaminocarbonyl, aminoalkyl-aminocarbonyl, aminoalkyl-carboxyamido, a quaternary ammonium alkylcarboxy, di(alkyl)-aminoalkyl, H.sub.2N—HC(Q.sub.5)-C(O)—O—, H.sub.2N—HC(Q.sub.5)-C(O)—N(H)—, azidoalkyloxy, cyanoalkyloxy, P.G.-HN—HC(Q.sub.5)-C(O)—O—, guanidinealkyloxy, and guanidinoalkylcarboxy.

[0102] In some embodiments, R.sub.1 through R.sub.4, R.sub.6, R.sub.7, R.sub.11, R.sub.12, R.sub.15, R.sub.16, and R.sub.18 are independently selected from the group consisting of hydrogen, hydroxyl, (C.sub.1-C.sub.22) alkyl, (C.sub.1-C.sub.22) hydroxyalkyl, (C.sub.1-C.sub.22) alkyloxy-(C.sub.1-C.sub.22) alkyl, (C.sub.1-C.sub.22) alkylcarboxy-(C.sub.1-C.sub.22) alkyl, (C.sub.1-C.sub.22) alkylamino-(C.sub.1-C.sub.22) alkyl, (C.sub.1-C.sub.22) alkylamino-(C.sub.1-C.sub.22) alkylamino, (C.sub.1-C.sub.22) alkylamino-(C.sub.1-C.sub.22) alkylamino-(C.sub.1-C.sub.22) alkylamino, (C.sub.1-C.sub.22) aminoalkyl, aryl, arylamino-(C.sub.1-C.sub.22) alkyl, (C.sub.1-C.sub.22) haloalkyl, C.sub.2-C.sub.6 alkenyl, C.sub.2-C.sub.6 alkynyl, oxo, a linking group attached to a second steroid, (C.sub.1-C.sub.22) aminoalkyloxy, (C.sub.1-C.sub.22) aminoalkyloxy-(C.sub.1-C.sub.22) alkyl, (C.sub.1-C.sub.22) aminoalkylcarboxy, (C.sub.1-C.sub.22) aminoalkylaminocarbonyl, (C.sub.1-C.sub.22) aminoalkyl-carboxamido, di(C.sub.1-

C.sub.22 alkyl)aminoalkyl, H.sub.2N—HC(Q.sub.5)—C(O)—O—, H.sub.2N—HC(Q.sub.5)—C(O)—N(H)—, (C.sub.1-C.sub.22) azidoalkyloxy, (C.sub.1-C.sub.22) cyanoalkyloxy, P.G.-HN—HC(Q.sub.5)—C(O)—O—, (C.sub.1-C.sub.22) guanidinoalkyloxy, (C.sub.1-C.sub.22) quaternary ammonium alkylcarboxy, and (C.sub.1-C.sub.22) guanidinoalkyl carboxy, where Q.sub.5 is a side chain of an amino acid (including a side chain of glycine, i.e., H), and P.G. is an amino protecting group; and [0103] R.sub.5, R.sub.8, R.sub.9, R.sub.10, R.sub.13, R.sub.14 and R.sub.17 are independently deleted when one of rings A, B, C, or D is unsaturated so as to complete the valency of the carbon atom at that site, or R.sub.5, R.sub.8, R.sub.9, R.sub.10, R.sub.13, and R.sub.14 are independently selected from the group consisting of hydrogen, hydroxyl, (C.sub.1-C.sub.22) alkyl, (C.sub.1-C.sub.22) hydroxyalkyl, (C.sub.1-C.sub.22) alkyloxy-(C.sub.1-C.sub.22) alkyl, (C.sub.1-C.sub.22) aminoalkyl, aryl, (C.sub.1-C.sub.22) haloalkyl, (C.sub.2-C.sub.6) alkenyl, (C.sub.2-C.sub.6) alkynyl, oxo, a linking group attached to a second steroid, (C.sub.1-C.sub.22) aminoalkyloxy, (C.sub.1-C.sub.22) aminoalkylcarboxy, (C.sub.1-C.sub.22) aminoalkylaminocarbonyl, di(C.sub.1-C.sub.22 alkyl)aminoalkyl, H.sub.2N—HC(Q.sub.5)—C(O)—O—, H.sub.2N—HC(Q.sub.5)—C(O)—N(H)—, (C.sub.1-C.sub.22) azidoalkyloxy, (C.sub.1-C.sub.22) cyanoalkyloxy, P.G.-HN—HC(Q.sub.5)—C(O)—O—, (C.sub.1-C.sub.22) guanidinoalkyloxy, and (C.sub.1-C.sub.22) guanidinoalkylcarboxy, where Q.sub.5 is a side chain of any amino acid, and P.G. is an amino protecting group; [0104] provided that at least two or three of R.sub.1-4, R.sub.6, R.sub.7, R.sub.1, R.sub.12, R.sub.15, R.sub.16, R.sub.17, and R.sub.18 are independently selected from the group consisting of (C.sub.1-C.sub.22) aminoalkyl, (C.sub.1-C.sub.22) aminoalkyloxy, (C.sub.1-C.sub.22) alkylcarboxy-(C.sub.1-C.sub.22) alkyl, (C.sub.1-C.sub.22) alkylamino-(C.sub.1-C.sub.22) alkylamino, (C.sub.1-C.sub.22) alkylamino-(C.sub.1-C.sub.22) alkylamino (C.sub.1-C.sub.22) alkylamino, (C.sub.1-C.sub.22) aminoalkylcarboxy, arylamino (C.sub.1-C.sub.22) alkyl, (C.sub.1-C.sub.22) aminoalkyloxy (C.sub.1-C.sub.22) aminoalkylaminocarbonyl, (C.sub.1-C.sub.22) aminoalkylaminocarbonyl, (C.sub.1-C.sub.22) aminoalkylcarboxy-amido, (C.sub.1-C.sub.22) quaternary ammonium alkylcarboxy, di(C.sub.1-C.sub.22 alkyl)aminoalkyl, H.sub.2N—HC(Q.sub.5)—C(O)—O—, H.sub.2N—HC(Q.sub.5)—C(O)—N(H)—, (C.sub.1-C.sub.22) azidoalkyloxy, (C.sub.1-C.sub.22) cyanoalkyloxy, P.G.-HN—HC(Q.sub.5)—C(O)—O—, (C.sub.1-C.sub.22) guanidinoalkyloxy, and (C.sub.1-C.sub.22) guanidinoalkylcarboxy.

[0105] In some embodiments, R.sub.1 through R.sub.4, R.sub.6, R.sub.7, R.sub.1, R.sub.12, R.sub.15, R.sub.16, and R.sub.18 are independently selected from the group consisting of hydrogen, hydroxyl, an unsubstituted (C.sub.1-C.sub.18) alkyl, unsubstituted (C.sub.1-C.sub.18) hydroxyalkyl, unsubstituted (C.sub.1-C.sub.18) alkyloxy-(C.sub.1-C.sub.18) alkyl, unsubstituted (C.sub.1-C.sub.18) alkylcarboxy-(C.sub.1-C.sub.18) alkyl, unsubstituted (C.sub.1-C.sub.18) alkylamino-(C.sub.1-C.sub.18) alkyl, unsubstituted (C.sub.1-C.sub.18) alkylamino-(C.sub.1-C.sub.18) alkylamino, (C.sub.1-C.sub.18) alkylamino-(C.sub.1-C.sub.18) alkylamino-(C.sub.1-C.sub.18) alkylamino, an unsubstituted (C.sub.1-C.sub.18) aminoalkyl, an unsubstituted aryl, an unsubstituted arylamino-(C.sub.1-C.sub.18) alkyl, oxo, an unsubstituted (C.sub.1-C.sub.18) aminoalkyloxy, an unsubstituted (C.sub.1-C.sub.18) aminoalkyloxy-(C.sub.1-C.sub.18) alkyl, an unsubstituted (C.sub.1-C.sub.18) aminoalkylcarboxy, an unsubstituted (C.sub.1-C.sub.18) aminoalkylaminocarbonyl, an unsubstituted (C.sub.1-C.sub.18) aminoalkyl-carboxamido, an unsubstituted di(C.sub.1-C.sub.18 alkyl)aminoalkyl, unsubstituted (C.sub.1-C.sub.18) guanidinoalkyloxy, unsubstituted (C.sub.1-C.sub.18) quaternary ammonium alkylcarboxy, and unsubstituted (C.sub.1-C.sub.18) guanidinoalkyl carboxy; and

[0106] R.sub.5, R.sub.8, R.sub.9, R.sub.10, R.sub.13, R.sub.14 and R.sub.17 are independently deleted when one of rings A, B, C, or D is unsaturated so as to complete the valency of the carbon atom at that site, or R.sub.5, R.sub.8, R.sub.9, R.sub.10, R.sub.13, and R 14 are independently selected from the group consisting of hydrogen, hydroxyl, an unsubstituted (C.sub.1-C.sub.18) alkyl, unsubstituted (C.sub.1-C.sub.18) hydroxyalkyl, unsubstituted (C.sub.1-C.sub.18) alkyloxy-

(C.sub.1-C.sub.18) alkyl, unsubstituted (C.sub.1-C.sub.18) alkylcarboxy-(C.sub.1-C.sub.18) alkyl, unsubstituted (C.sub.1-C.sub.18) alkylamino-(C.sub.1-C.sub.18) alkyl, (C.sub.1-C.sub.18) alkylamino-(C.sub.1-C.sub.18) alkylamino, unsubstituted (C.sub.1-C.sub.18) alkylamino-(C.sub.1-C.sub.18) alkylamino-(C.sub.1-C.sub.18) alkylamino, an unsubstituted (C.sub.1-C.sub.18) aminoalkyl, an unsubstituted aryl, an unsubstituted arylamino-(C.sub.1-C.sub.18) alkyl, oxo, an unsubstituted (C.sub.1-C.sub.18) aminoalkyloxy, an unsubstituted (C.sub.1-C.sub.18) aminoalkyloxy-(C.sub.1-C.sub.18) alkyl, an unsubstituted (C.sub.1-C.sub.18) aminoalkylcarboxy, an unsubstituted (C.sub.1-C.sub.18) aminoalkyl-aminocarbonyl, an unsubstituted (C.sub.1-C.sub.18) aminoalkylcarboxamido, an unsubstituted di(C.sub.1-C.sub.18 alkyl)aminoalkyl, unsubstituted (C.sub.1-C.sub.18) guanidinoalkyloxy, unsubstituted (C.sub.1-C.sub.18) quaternary ammonium alkylcarboxy, and unsubstituted (C.sub.1-C.sub.18) guanidinoalkyl carboxy, [0107] provided that at least two or three of R.sub.1-4, R.sub.6, R.sub.7, R.sub.11, R.sub.12, R.sub.15, R.sub.16, R.sub.17, and R.sub.18 are independently selected from the group consisting of hydrogen, hydroxyl, an unsubstituted (C.sub.1-C.sub.18) alkyl, unsubstituted (C.sub.1-C.sub.18) hydroxyalkyl, unsubstituted (C.sub.1-C.sub.18) alkyloxy-(C.sub.1-C.sub.18) alkyl, unsubstituted (C.sub.1-C.sub.18) alkylcarboxy-(C.sub.1-C.sub.18) alkyl, unsubstituted (C.sub.1-C.sub.18) alkylamino-(C.sub.1-C.sub.18) alkyl, unsubstituted (C.sub.1-C.sub.18) alkylamino-(C.sub.1-C.sub.18) alkylamino, unsubstituted (C.sub.1-C.sub.18) alkylamino-(C.sub.1-C.sub.18) alkylamino-(C.sub.1-C.sub.18) aminoalkyl, an unsubstituted aryl, an unsubstituted arylamino-(C.sub.1-C.sub.18) alkyl, oxo, an unsubstituted (C.sub.1-C.sub.18) aminoalkyloxy, an unsubstituted (C.sub.1-C.sub.18) aminoalkyloxy-(C.sub.1-C.sub.18) alkyl, an unsubstituted (C.sub.1-C.sub.18) aminoalkylcarboxy, an unsubstituted (C.sub.1-C.sub.18) aminoalkylaminocarbonyl, an unsubstituted [0108] (C.sub.1-C.sub.18) aminoalkylcarboxamido, an unsubstituted di(C.sub.1-C.sub.18 alkyl)aminoalkyl, unsubstituted (C.sub.1-C.sub.18) guanidinoalkyloxy, unsubstituted (C.sub.1-C.sub.18) quaternary ammonium alkylcarboxy, and unsubstituted (C.sub.1-C.sub.18) guanidinoalkyl carboxy.

[0109] In some embodiments, R.sub.3, R.sub.7, R.sub.12, and R.sub.18 are independently selected from the group consisting of hydrogen, an unsubstituted (C.sub.1-C.sub.18) alkyl, unsubstituted (C.sub.1-C.sub.18) hydroxyalkyl, unsubstituted (C.sub.1-C.sub.18) alkyloxy-(C.sub.1-C.sub.18) alkyl, unsubstituted (C.sub.1-C.sub.18) alkylcarboxy-(C.sub.1-C.sub.18) alkyl, unsubstituted (C.sub.1-C.sub.18) alkylamino-(C.sub.1-C.sub.18) alkyl, unsubstituted (C.sub.1-C.sub.18) alkylamino-(C.sub.1-C.sub.18) alkylamino, unsubstituted (C.sub.1-C.sub.18) alkylamino-(C.sub.1-C.sub.18) aminoalkyl, an unsubstituted arylamino-(C.sub.1-C.sub.18) alkyl, an unsubstituted (C.sub.1-C.sub.18) aminoalkyloxy, an unsubstituted (C.sub.1-C.sub.18) aminoalkyloxy-(C.sub.1-C.sub.18) alkyl, an unsubstituted (C.sub.1-C.sub.18) aminoalkylcarboxy, an unsubstituted (C.sub.1-C.sub.18) aminoalkyl-aminocarbonyl, an unsubstituted (C.sub.1-C.sub.18) aminoalkylcarboxamido, an unsubstituted di-(C.sub.1-C.sub.18 alkyl)aminoalkyl, unsubstituted (C.sub.1-C.sub.18) guanidinoalkyloxy, unsubstituted (C.sub.1-C.sub.18) quaternary ammonium alkylcarboxy, and unsubstituted (C.sub.1-C.sub.18) guanidinoalkyl carboxy.

[0110] In some embodiments, R.sub.1, R.sub.2, R.sub.4, R.sub.5, R.sub.6, R.sub.8, R.sub.9, R.sub.10, R.sub.11, R.sub.13, R.sub.14, R.sub.15, R.sub.16, and R.sub.17 are independently selected from the group consisting of hydrogen and unsubstituted (C.sub.1-C.sub.6) alkyl.

[0111] In some embodiments, R.sub.3, R.sub.7, R.sub.12, and R.sub.18 are independently selected from the group consisting of hydrogen, an unsubstituted (C.sub.1-C.sub.6) alkyl, unsubstituted (C.sub.1-C.sub.6) hydroxyalkyl, unsubstituted (C.sub.1-C.sub.16) alkyloxy-(C.sub.1-C.sub.5) alkyl, unsubstituted (C.sub.1-C.sub.16) alkylcarboxy-(C.sub.1-C.sub.5) alkyl, unsubstituted (C.sub.1-C.sub.16) alkylamino-(C.sub.1-C.sub.5) alkyl, (C.sub.1-C.sub.16) alkylamino-(C.sub.1-C.sub.5) alkylamino, unsubstituted (C.sub.1-C.sub.16) alkylamino-(C.sub.1-C.sub.16) alkylamino-(C.sub.1-C.sub.5) alkylamino, an unsubstituted (C.sub.1-C.sub.16) aminoalkyl, an unsubstituted

aryl-amino-(C.sub.1-C.sub.5) alkyl, an unsubstituted (C.sub.1-C.sub.5) aminoalkyloxy, an unsubstituted (C.sub.1-C.sub.16) aminoalkyloxy-(C.sub.1-C.sub.5) alkyl, an unsubstituted (C.sub.1-C.sub.5) aminoalkylcarboxy, an unsubstituted (C.sub.1-C.sub.5) aminoalkylaminocarbonyl, an unsubstituted (C.sub.1-C.sub.5) aminoalkylcarboxamido, an unsubstituted di(C.sub.1-C.sub.5 alkyl)amino-(C.sub.1-C.sub.5) alkyl, unsubstituted (C.sub.1-C.sub.5) guanidinoalkyloxy, unsubstituted (C.sub.1-C.sub.16) quaternary ammonium alkylcarboxy, and unsubstituted (C.sub.1-C.sub.16) guanidinoalkylcarboxy.

[0112] In some embodiments, R.sub.1, R.sub.2, R.sub.4, R.sub.5, R.sub.6, R.sub.8, R.sub.10, R.sub.11, R.sub.14, R.sub.16, and R.sub.17 are each hydrogen; and R.sub.9 and R.sub.13 are each methyl.

[0113] In some embodiments, R.sub.3, R.sub.7, R.sub.12, and R.sub.18 are independently selected from the group consisting of aminoalkyloxy; aminoalkylcarboxy; alkylaminoalkyl; alkoxy-carbonylalkyl; alkylcarbonylalkyl; di(alkyl)aminoalkyl; alkylcarboxyalkyl; and hydroxyalkyl.

[0114] In some embodiments, R.sub.3, R.sub.7, and R.sub.12 are independently selected from the group consisting of aminoalkyloxy and aminoalkylcarboxy; and R.sub.18 is selected from the group consisting of alkylaminoalkyl; alkoxy-carbonylalkyl; alkylcarbonyloxyalkyl; di(alkyl)-aminoalkyl; alkylaminoalkyl; alkoxy-carbonylalkyl; alkylcarboxyalkyl; and hydroxyalkyl.

[0115] In some embodiments, R.sub.3, R.sub.7, and R.sub.12 are the same.

[0116] In some embodiments, R.sub.3, R.sub.7, and R.sub.12 are aminoalkyloxy.

[0117] In some embodiments, R.sub.18 is alkylaminoalkyl.

[0118] In some embodiments, R.sub.18 is alkoxy-carbonylalkyl.

[0119] In some embodiments, R.sub.18 is di(alkyl)aminoalkyl.

[0120] In some embodiments, R.sub.18 is alkylcarboxyalkyl.

[0121] In some embodiments, R.sub.18 is hydroxyalkyl.

[0122] In some embodiments, R.sub.3, R.sub.7, and R.sub.12 are aminoalkylcarboxy.

[0123] In some embodiments, R.sub.3, R.sub.7, R.sub.12, and R.sub.18 are independently selected from the group consisting of aminoalkyloxy; aminoalkylcarboxy; alkylaminoalkyl; di(alkyl)aminoalkyl; alkoxy-carbonylalkyl; and alkylcarboxyalkyl.

[0124] In some embodiments, R.sub.3, R.sub.7, R.sub.12, and R.sub.18 are independently selected from the group consisting of aminoalkyloxy; aminoalkylcarboxy; alkylaminoalkyl; di(alkyl)aminoalkyl; and alkoxy-carbonylalkyl.

[0125] In some embodiments, R.sub.3, R.sub.7, and R.sub.12 are independently selected from the group consisting of aminoalkyloxy and aminoalkylcarboxy, and wherein R.sub.18 is selected from the group consisting of alkylaminoalkyl; di(alkyl)aminoalkyl; alkoxy-carbonylalkyl; and alkylcarboxyalkyl.

[0126] In some embodiments, R.sub.3, R.sub.7, and R.sub.12 are independently selected from the group consisting of aminoalkyloxy and aminoalkylcarboxy, and wherein R.sub.18 is selected from the group consisting of alkylaminoalkyl; di(alkyl)aminoalkyl; and alkoxy-carbonylalkyl.

[0127] In some embodiments, R.sub.3, R.sub.7, R.sub.12, and R.sub.18 are independently selected from the group consisting of amino-C.sub.3-alkyloxy; amino-C.sub.3-alkyl-carboxy; C.sub.8-alkylamino-C.sub.5-alkyl; C.sub.12-alkylamino-C.sub.5-alkyl; C.sub.13-alkylamino-C.sub.5-alkyl; C.sub.16-alkylamino-C.sub.5-alkyl; di-(C.sub.5-alkyl)amino-C.sub.5-alkyl; C.sub.6-alkoxy-carbonyl-C.sub.4-alkyl; C.sub.8-alkoxy-carbonyl-C.sub.4-alkyl; C.sub.10-alkoxy-carbonyl-C.sub.4-alkyl; C.sub.6-alkyl-carboxy-C.sub.4-alkyl; C.sub.a-alkyl-carboxy-C.sub.4-alkyl; and C.sub.10-alkyl-carboxy-C.sub.4-alkyl.

[0128] In some embodiments, R.sub.3, R.sub.7, R.sub.12, and R.sub.18 are independently selected from the group consisting of amino-C.sub.3-alkyloxy; amino-C.sub.3-alkyl-carboxy; C.sub.8-alkylamino-C.sub.5-alkyl; C.sub.12-alkylamino-C.sub.5-alkyl; C.sub.13-alkylamino-C.sub.5-alkyl; C.sub.16-alkylamino-C.sub.5-alkyl; di-(C.sub.5-alkyl)amino-C.sub.5-alkyl; C.sub.6-alkoxy-

carbonyl-C.sub.4-alkyl; C.sub.8-alkoxy-carbonyl-C.sub.4-alkyl; and C.sub.10-alkoxy-carbonyl-C.sub.4-alkyl.

[0129] In some embodiments, R.sub.3, R.sub.7, and R.sub.12, are independently selected from the group consisting of amino-C.sub.3-alkyloxy or amino-C.sub.3-alkyl-carboxy, and wherein R.sub.18 is selected from the group consisting of C.sub.8-alkylamino-C.sub.5-alkyl; C.sub.12-alkylamino-C.sub.5-alkyl; C.sub.13-alkylamino-C.sub.5-alkyl; C.sub.16-alkylamino-C.sub.5-alkyl; di-(C.sub.5-alkyl)amino-C.sub.5-alkyl; C.sub.6-alkoxy-carbonyl-C.sub.4-alkyl; C.sub.8-alkoxy-carbonyl-C.sub.4-alkyl; C.sub.10-alkoxy-carbonyl-C.sub.4-alkyl; C.sub.6-alkyl-carboxy-C.sub.4-alkyl; C.sub.8-alkyl-carboxy-C.sub.4-alkyl; and C.sub.10-alkyl-carboxy-C.sub.4-alkyl.

[0130] In some embodiments, R.sub.3, R.sub.7, and R.sub.12, are independently selected from the group consisting of amino-C.sub.3-alkyloxy or amino-C.sub.3-alkyl-carboxy, and wherein R.sub.18 is selected from the group consisting of C.sub.8-alkylamino-C.sub.5-alkyl; C.sub.12-alkylamino-C.sub.5-alkyl; C.sub.13-alkylamino-C.sub.5-alkyl; C.sub.16-alkylamino-C.sub.5-alkyl; di-(C.sub.5-alkyl)amino-C.sub.5-alkyl; C.sub.6-alkoxy-carbonyl-C.sub.4-alkyl; C.sub.8-alkoxy-carbonyl-C.sub.4-alkyl; and C.sub.10-alkoxy-carbonyl-C.sub.4-alkyl.

[0131] In some embodiments, R.sub.3, R.sub.7, R.sub.12, and R.sub.18 are independently selected from the group consisting of amino-C.sub.3-alkyloxy; amino-C.sub.3-alkyl-carboxy; amino-C.sub.2-alkylcarboxy; C.sub.8-alkylamino-C.sub.5-alkyl; C.sub.8-alkoxy-carbonyl-C.sub.4-alkyl; C.sub.10-alkoxy-carbonyl-C.sub.4-alkyl; C.sub.8-alkyl-carbonyl-C.sub.4-alkyl; di-(C.sub.5-alkyl)amino-C.sub.5-alkyl; C.sub.13-alkylamino-C.sub.5-alkyl; C.sub.6-alkoxy-carbonyl-C.sub.4-alkyl; C.sub.6-alkyl-carboxy-C.sub.4-alkyl; C.sub.16-alkylamino-C.sub.5-alkyl; C.sub.12-alkylamino-C.sub.5-alkyl; and hydroxy(C.sub.5) alkyl.

[0132] In some embodiments, R.sub.18 is selected from the group consisting of C.sub.8-alkylamino-C.sub.5-alkyl or C.sub.8-alkoxy-carbonyl-C.sub.4-alkyl.

[0133] In some embodiments, one or more of rings A, B, C, and D are heterocyclic.

[0134] In some embodiments, rings A, B, C, and D are non-heterocyclic.

[0135] In some embodiments, the CSA compound is a compound of Formula (III), or salt thereof, having a steroidal backbone:

##STR00006##

[0136] In some embodiments, R.sub.3, R.sub.7, and R.sub.12 are independently selected from the group consisting of hydrogen, an unsubstituted (C.sub.1-C.sub.22) alkyl, unsubstituted (C.sub.1-C.sub.22) hydroxyalkyl, unsubstituted (C.sub.1-C.sub.22) alkyloxy-(C.sub.1-C.sub.22) alkyl, unsubstituted (C.sub.1-C.sub.22) alkylcarboxy-(C.sub.1-C.sub.22) alkyl, unsubstituted (C.sub.1-C.sub.22) alkylamino-(C.sub.1-C.sub.22) alkyl, unsubstituted (C.sub.1-C.sub.22) alkylamino-(C.sub.1-C.sub.22) alkylamino, unsubstituted (C.sub.1-C.sub.22) alkylamino-(C.sub.1-C.sub.22) alkylamino-(C.sub.1-C.sub.18) alkylamino, an unsubstituted (C.sub.1-C.sub.22) aminoalkyl, an unsubstituted arylamino-(C.sub.1-C.sub.22) alkyl, an unsubstituted (C.sub.1-C.sub.22) aminoalkyloxy, an unsubstituted (C.sub.1-C.sub.22) aminoalkyloxy-(C.sub.1-C.sub.22) alkyl, an unsubstituted (C.sub.1-C.sub.22) aminoalkylcarboxy, an unsubstituted (C.sub.1-C.sub.22) aminoalkyl-aminocarbonyl, an unsubstituted (C.sub.1-C.sub.22) aminoalkylcarboxamido, an unsubstituted di(C.sub.1-C.sub.22 alkyl)aminoalkyl, unsubstituted (C.sub.1-C.sub.22) guanidinoalkyloxy, unsubstituted (C.sub.1-C.sub.22) quaternary ammonium alkylcarboxy, and unsubstituted (C.sub.1-C.sub.22) guanidinoalkyl carboxy.

[0137] In some embodiments, R.sub.3, R.sub.7, and R.sub.12 are independently selected from the group consisting of hydrogen, an unsubstituted (C.sub.1-C.sub.6) alkyl, unsubstituted (C.sub.1-C.sub.6) hydroxyalkyl, unsubstituted (C.sub.1-C.sub.16) alkyloxy-(C.sub.1-C.sub.5) alkyl, unsubstituted (C.sub.1-C.sub.16) alkylcarboxy-(C.sub.1-C.sub.5) alkyl, unsubstituted (C.sub.1-C.sub.16) alkylamino-(C.sub.1-C.sub.5) alkyl, unsubstituted (C.sub.1-C.sub.16) alkylamino-(C.sub.1-C.sub.5) alkylamino, unsubstituted (C.sub.1-C.sub.16) alkylamino-(C.sub.1-C.sub.16) alkylamino-(C.sub.1-C.sub.5) alkylamino, an unsubstituted (C.sub.1-C.sub.16) aminoalkyl, an

unsubstituted arylamino-(C.sub.1-C.sub.5) alkyl, an unsubstituted (C.sub.1-C.sub.5) aminoalkoxy, an unsubstituted (C.sub.1-C.sub.16) aminoalkoxy-(C.sub.1-C.sub.5) alkyl, an unsubstituted (C.sub.1-C.sub.5) aminoalkylcarboxy, an unsubstituted (C.sub.1-C.sub.5) aminoalkylaminocarbonyl, an unsubstituted (C.sub.1-C.sub.5) aminoalkylcarboxamido, an unsubstituted di(C.sub.1-C.sub.5 alkyl)amino-(C.sub.1-C.sub.5) alkyl, unsubstituted (C.sub.1-C.sub.5) guanidinoalkoxy, unsubstituted (C.sub.1-C.sub.16) quaternary ammonium alkylcarboxy, and unsubstituted (C.sub.1-C.sub.16) guanidinoalkylcarboxy.

[0138] In some embodiments, R.sub.3, R.sub.7, and R.sub.12 are independently selected from the group consisting of aminoalkoxy; aminoalkylcarboxy; alkylaminoalkyl; alkoxy-carbonylalkyl; alkyl-carbonylalkyl; di(alkyl)aminoalkyl; alkylcarboxyalkyl; and hydroxyalkyl.

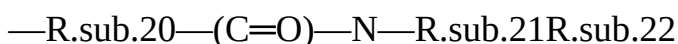
[0139] In some embodiments, R.sub.3, R.sub.7, and R.sub.12 are independently selected from the group consisting of aminoalkoxy and aminoalkylcarboxy.

[0140] In some embodiments, R.sub.3, R.sub.7, and R.sub.12 are the same. In some embodiments, R.sub.3, R.sub.7, and R.sub.12 are aminoalkoxy. In some embodiments, R.sub.3, R.sub.7, and R.sub.12 are aminoalkylcarboxy.

[0141] In some embodiments, R.sub.3, R.sub.7, and R.sub.12 are independently selected from the group consisting of amino-C.sub.3-alkoxy; amino-C.sub.3-alkyl-carboxy; C.sub.8-alkylamino-C.sub.5-alkyl; C.sub.8-alkoxy-carbonyl-C.sub.4-alkyl; C.sub.8-alkyl-carbonyl-C.sub.4-alkyl; di-(C.sub.5-alkyl)amino-C.sub.5-alkyl; C.sub.13-alkylamino-C.sub.5-alkyl; C.sub.6-alkoxy-carbonyl-C.sub.4-alkyl; C.sub.6-alkyl-carboxy-C.sub.4-alkyl; and C.sub.16-alkylamino-C.sub.5-alkyl.

[0142] In some embodiments, CSA compounds as disclosed herein can be a compound of Formula (I), Formula (II), Formula (III), or salts thereof wherein at least R.sub.18 of the steroidal backbone includes amide functionality in which the carbonyl group of the amide is positioned between the amido nitrogen of the amide and fused ring D of the steroidal backbone. For example, any of the embodiments described above can substitute R.sub.18 for an R.sub.18 including amide functionality in which the carbonyl group of the amide is positioned between the amido nitrogen of the amide and fused ring D of the steroidal backbone.

[0143] In some embodiments, at least R.sub.18 can have the following structure:



wherein R.sub.20 is omitted or alkyl, alkenyl, alkynyl, or aryl, and R.sub.21 and R.sub.22 are independently selected from the group consisting of hydrogen, alkyl, alkenyl, alkynyl, or aryl, provided that at least one of R.sub.21 and R.sub.22 is not hydrogen.

[0144] In some embodiments, R.sub.21 and R.sub.22 are independently selected from the group consisting of hydrogen, C.sub.1-C.sub.24 alkyl, C.sub.2-C.sub.24 alkenyl, C.sub.2-C.sub.24 alkynyl, C.sub.6 or C.sub.10 aryl, 5 to 10 membered heteroaryl, 5 to 10 membered heterocyclyl, C.sub.7-13 aralkyl, (5 to 10 membered heteroaryl)---C.sub.1-C.sub.6 alkyl, C.sub.3-10 carbocyclyl, C.sub.4-10 (carbocyclyl)alkyl, (5 to 10 membered heterocyclyl)---C.sub.1-C.sub.6 alkyl, amido, and a suitable amine protecting group, provided that at least one of R.sub.21 and R.sub.22 is not hydrogen. In some embodiments, R.sub.21 and R.sub.22, together with the atoms to which they are attached, form a 5 to 10 membered heterocyclyl ring.

[0145] In some embodiments, the CSA is selected from the group consisting of:

##STR00007## ##STR00008## ##STR00009##

[0146] In some embodiments, the CSA is

##STR00010##

[0147] In some embodiments, the CSA is

##STR00011##

[0148] In some embodiments, the CSA is

##STR00012##

[0149] In some embodiments, the CSA is

##STR00013##

[0150] In some embodiments, the CSA is

##STR00014##

[0151] In some embodiments, the CSA is

##STR00015##

[0152] In some embodiments, the CSA is

##STR00016##

[0153] In some embodiments, the CSA is

##STR00017##

[0154] In some embodiments, the CSA is

##STR00018##

[0155] In some embodiments, the CSA is

##STR00019##

[0156] In some embodiments, the CSA is

##STR00020##

[0157] In some embodiments, the CSA is

##STR00021##

CSA Salts

[0158] It has been discovered that the CSA salt form can be manipulated by the choice of counterion to afford CSA salts having pharmaceutically beneficial properties such as improved solubility, crystallinity, flow, and storage stability. Such properties are of critical concern for the handling and use of CSAs as pharmaceutical agents. For example, poor solubility can influence the ultimate formulation of a CSA, while storage stability can influence efficient manufacturing protocols and shelf life of the CSA formulation. Moreover, crystallinity of the CSA can affect purification and significantly influence the synthesis and handling of the CSA during manufacturing. Likewise, the flow properties of a CSA can influence the equipment and handling of a CSA during manufacturing. Thus, the ability to manipulate and control these properties through the selection of an appropriate counterion represents a significant step toward the commercialization of a CSA pharmaceutical product.

[0159] Some embodiments are directed to a sulfuric acid addition salt or sulfonic acid addition salt of a CSA. In some embodiments, the sulfonic acid addition salt is a disulfonic acid addition salt. In some embodiments, the sulfonic acid addition salt is a 1,5-naphthalenedisulfonic acid addition salt. In some embodiments, the acid addition salt is a mono-addition salt. In other embodiments, the acid addition salt is a di-addition salt. In other embodiments, the acid addition salt is a tetra-addition salt.

[0160] In some embodiments, the acid addition salt described above is a solid.

[0161] In some embodiments, the acid addition salt described above is a flowable solid.

[0162] In some embodiments, the acid addition salt described above is crystalline.

[0163] In some embodiments, the acid addition salt described above is storage stable. In some embodiments, the acid addition salt is storage stable for a period of 5 days, 1 week, 2 weeks, 1 month, 3 months, 6 months, 1 year, or about any of the aforementioned numbers, or a range bounded by any two of the aforementioned numbers. In some embodiments, storage stability is measured by degradation that is less than 0.5%, 1%, 2%, 3%, 4%, 5%, 10% or about any of the aforementioned numbers, or a range bounded by any two of the aforementioned numbers for a given period of time, as described above. In some embodiments, storage stability is measured qualitatively by a change in crystallinity, such as loss of crystallinity and/or the concomitant increase in amorphous materials such as amorphous solids, gums, and the like, for a given period of time, as described above.

CSA Salts Synthesis

[0164] Some embodiments are directed to a process for preparing a CSA acid addition salt, in

which 1-4 equivalents of sulfuric acid or a sulfonic acid is contacted with a CSA. In some embodiments, the sulfonic acid addition salt is a disulfonic acid addition salt. In some embodiments, the sulfonic acid addition salt is a 1,5-naphthalenedisulfonic acid (NDSA) addition salt. In some embodiments, the acid addition salt is a mono-addition salt. In other embodiments, the acid addition salt is a di-addition salt. In other embodiments, the acid addition salt is a tetra-addition salt. In some embodiments, 1, 2, 3, or 4 equivalents of acid, or about any of the aforementioned numbers, or a range bounded by any of the aforementioned numbers is contacted with the CSA.

[0165] In some embodiments, the process for preparing the above-described CSA salt includes diluting the free base of a CSA with a solvent; adding at least one equivalent of an acid to the diluted CSA in solvent to afford a reaction mixture; precipitating or temperature cycling the reaction mixture; and isolating a CSA salt. In some embodiments, the CSA salt is precipitated. In other embodiments, the CSA salt is isolated after temperature cycling. In some embodiments, the temperature cycling is conducted for at least about 1, 2, 3, 6, 8, 12, 16, 18, 20, 24, 36, or 48 hours, or a range bounded by any two of the aforementioned numbers. In some embodiments, the CSA salt is isolated after the addition of an anti-solvent. In other embodiments, the CSA salt is isolated after evaporation of solvent.

Pharmaceutical Compositions

[0166] While it is possible for the compounds described herein to be administered alone, it may be preferable to formulate the compounds as pharmaceutical compositions (i.e., formulations). As such, in yet another aspect, pharmaceutical compositions useful in the methods and uses of the disclosed embodiments are provided. A pharmaceutical composition is any composition that may be administered in vitro or in vivo or both to a subject in order to treat or ameliorate a condition. In a preferred embodiment, a pharmaceutical composition may be administered in vivo. A subject may include one or more cells or tissues, or organisms. In some exemplary embodiments, the subject is an animal. In some embodiments, the animal is a mammal. The mammal may be a human or primate in some embodiments. A mammal includes any mammal, such as by way of non-limiting example, cattle, pigs, sheep, goats, horses, camels, buffalo, cats, dogs, rats, mice, and humans.

[0167] As used herein the terms “pharmaceutically acceptable” and “physiologically acceptable” mean a biologically compatible formulation, gaseous, liquid or solid, or mixture thereof, which is suitable for one or more routes of administration, in vivo delivery, or contact. A formulation is compatible in that it does not destroy activity of an active ingredient therein (e.g., a CSA compound), or induce adverse side effects that far outweigh any prophylactic or therapeutic effect or benefit.

[0168] In some embodiments, pharmaceutical compositions may be formulated with pharmaceutically acceptable excipients such as carriers, solvents, stabilizers, adjuvants, diluents, etc., depending upon the particular mode of administration and dosage form. The pharmaceutical compositions should generally be formulated to achieve a physiologically compatible pH, and may range from a pH of about 3 to a pH of about 11, preferably about pH 3 to about pH 7, depending on the formulation and route of administration. In alternative embodiments, it may be preferred that the pH is adjusted to a range from about pH 5.0 to about pH 8. More particularly, the pharmaceutical compositions may comprise a therapeutically or prophylactically effective amount of at least one compound as described herein, together with one or more pharmaceutically acceptable excipients. Optionally, the pharmaceutical compositions may comprise a combination of the compounds described herein, or may include a second active ingredient useful in the treatment or prevention of bacterial infection (e.g., anti-bacterial or anti-microbial agents). Optionally, the composition is formulated as a coating. In some embodiments, the coating is on a medical device. In some embodiments, the coating is on medical instrumentation.

[0169] Formulations, e.g., for parenteral or oral administration, are most typically solids, liquid solutions, emulsions or suspensions, while inhalable formulations for pulmonary administration are

generally liquids or powders, with powder formulations being generally preferred. A preferred pharmaceutical composition may also be formulated as a lyophilized solid that is reconstituted with a physiologically compatible solvent prior to administration. Alternative pharmaceutical compositions may be formulated as syrups, creams, ointments, tablets, and the like.

[0170] Compositions may contain one or more excipients. Pharmaceutically acceptable excipients are determined in part by the particular composition being administered, as well as by the particular method used to administer the composition. Accordingly, there exists a wide variety of suitable formulations of pharmaceutical compositions (see, e.g., Remington's Pharmaceutical Sciences).

[0171] Suitable excipients may be carrier molecules that include large, slowly metabolized macromolecules such as proteins, polysaccharides, polylactic acids, polyglycolic acids, polymeric amino acids, amino acid copolymers, and inactive virus particles. Other exemplary excipients include antioxidants such as ascorbic acid; chelating agents such as EDTA; carbohydrates such as dextrin, hydroxyalkylcellulose, hydroxyalkylmethylcellulose, stearic acid; liquids such as oils, water, saline, glycerol and ethanol; wetting or emulsifying agents; pH buffering substances; and the like. Liposomes are also included within the definition of pharmaceutically acceptable excipients.

[0172] Pharmaceutical compositions may be formulated in any form suitable for the intended method of administration. When intended for oral use for example, tablets, troches, lozenges, aqueous or oil suspensions, non-aqueous solutions, dispersible powders or granules (including micronized particles or nanoparticles), emulsions, hard or soft capsules, syrups or elixirs may be prepared. Compositions intended for oral use may be prepared according to any method known to the art for the manufacture of pharmaceutical compositions, and such compositions may contain one or more agents including sweetening agents, flavoring agents, coloring agents and preserving agents, in order to provide a palatable preparation.

[0173] Pharmaceutically acceptable excipients particularly suitable for use in conjunction with tablets include, for example, inert diluents, such as celluloses, calcium or sodium carbonate, lactose, calcium or sodium phosphate; disintegrating agents, such as cross-linked povidone, maize starch, or alginic acid; binding agents, such as povidone, starch, gelatin or acacia; and lubricating agents, such as magnesium stearate, stearic acid or talc.

[0174] Tablets may be uncoated or may be coated by known techniques including microencapsulation to delay disintegration and adsorption in the gastrointestinal tract and thereby provide a sustained action over a longer period. For example, a time delay material such as glyceryl monostearate or glyceryl distearate alone or with a wax may be employed.

[0175] Formulations for oral use may be also presented as hard gelatin capsules where the active ingredient is mixed with an inert solid diluent, for example celluloses, lactose, calcium phosphate or kaolin, or as soft gelatin capsules wherein the active ingredient is mixed with non-aqueous or oil medium, such as glycerin, propylene glycol, polyethylene glycol, peanut oil, liquid paraffin or olive oil.

[0176] In another embodiment, pharmaceutical compositions may be formulated as suspensions comprising a compound of the embodiments in admixture with at least one pharmaceutically acceptable excipient suitable for the manufacture of a suspension.

[0177] In yet another embodiment, pharmaceutical compositions may be formulated as dispersible powders and granules suitable for preparation of a suspension by the addition of suitable excipients.

[0178] Excipients suitable for use in connection with suspensions include suspending agents, such as sodium carboxymethylcellulose, methylcellulose, hydroxypropyl methylcellulose, sodium alginate, polyvinylpyrrolidone, gum tragacanth, gum acacia, dispersing or wetting agents such as a naturally occurring phosphatide (e.g., lecithin), a condensation product of an alkylene oxide with a fatty acid (e.g., polyoxyethylene stearate), a condensation product of ethylene oxide with a long chain aliphatic alcohol (e.g., heptadecaethyleneoxycethanol), a condensation product of ethylene oxide with a partial ester derived from a fatty acid and a hexitol anhydride (e.g., polyoxyethylene

sorbitan monooleate); polysaccharides and polysaccharide-like compounds (e.g. dextran sulfate); glycoaminoglycans and glycosaminoglycan-like compounds (e.g., hyaluronic acid); and thickening agents, such as carbomer, beeswax, hard paraffin or cetyl alcohol. The suspensions may also contain one or more preservatives such as acetic acid, methyl and/or n-propyl p-hydroxy-benzoate; one or more coloring agents; one or more flavoring agents; and one or more sweetening agents such as sucrose or saccharin.

[0179] Pharmaceutical compositions may also be in the form of oil-in water emulsions. The oily phase may be a vegetable oil, such as olive oil or arachis oil, a mineral oil, such as liquid paraffin, or a mixture of these. Suitable emulsifying agents include naturally-occurring gums, such as gum acacia and gum tragacanth; naturally occurring phosphatides, such as soybean lecithin, esters or partial esters derived from fatty acids; hexitol anhydrides, such as sorbitan monooleate; and condensation products of these partial esters with ethylene oxide, such as polyoxyethylene sorbitan monooleate. The emulsion may also contain sweetening and flavoring agents. Syrups and elixirs may be formulated with sweetening agents, such as glycerol, sorbitol or sucrose. Such formulations may also contain a demulcent, a preservative, a flavoring or a coloring agent.

[0180] Additionally, pharmaceutical compositions may be in the form of a sterile injectable preparation, such as a sterile injectable aqueous emulsion or oleaginous suspension. This emulsion or suspension may be formulated according to the known art using those suitable dispersing or wetting agents and suspending agents which have been mentioned above. The sterile injectable preparation may also be a sterile injectable solution or suspension in a non-toxic parenterally acceptable diluent or solvent, such as a solution in 1,2-propanediol.

[0181] Sterile injectable preparations may also be prepared as a lyophilized powder. Among the acceptable vehicles and solvents that may be employed are water, Ringer's solution, and isotonic sodium chloride solution. In addition, sterile fixed oils may be employed as a solvent or suspending medium. For this purpose, any bland fixed oil may be employed including synthetic mono- or diglycerides. In addition, fatty acids such as oleic acid may likewise be used in the preparation of injectables.

[0182] To obtain a stable water-soluble dose form of a pharmaceutical composition, a pharmaceutically acceptable salt of a compound described herein may be dissolved in an aqueous solution of an organic or inorganic acid, such as 0.3 M solution of succinic acid, or more preferably, citric acid. If a soluble salt form is not available, the compound may be dissolved in a suitable co-solvent or combination of co-solvents. Examples of suitable co-solvents include alcohol, propylene glycol, polyethylene glycol 300, polysorbate 80, glycerin and the like in concentrations ranging from about 0 to about 60% of the total volume. In one embodiment, the active compound is dissolved in DMSO and diluted with water.

[0183] Pharmaceutical composition may also be in the form of a solution of a salt form of the active ingredient in an appropriate aqueous vehicle, such as water or isotonic saline or dextrose solution. Also contemplated are compounds which have been modified by substitutions or additions of chemical or biochemical moieties which make them more suitable for delivery (e.g., increase solubility, bioactivity, palatability, decrease adverse reactions, etc.), for example by esterification, glycosylation, PEGylation, and complexation.

[0184] Many therapeutics have undesirably short half-lives and/or undesirable toxicity. Thus, the concept of improving half-life or toxicity is applicable to various treatments and fields.

Pharmaceutical compositions can be prepared, however, by complexing the therapeutic with a biochemical moiety to improve such undesirable properties. Proteins are a particular biochemical moiety that may be complexed with a CSA for administration in a wide variety of applications. In some embodiments, one or more CSAs are complexed with a protein. In some embodiments, one or more CSAs are complexed with a protein to increase the CSA's half-life. In other embodiments, one or more CSAs are complexed with a protein to decrease the CSA's toxicity. Albumin is a particularly preferred protein for complexation with a CSA. In some embodiments, the albumin is

fat-free albumin.

[0185] With respect to the CSA therapeutic, the biochemical moiety for complexation can be added to the pharmaceutical composition as 0.25, 0.5, 0.75, 1, 1.5, 2, 2.5, 3, 3.5, 4, 4.5, 5, 10, 20, 50, or 100 weight equivalents, or a range bounded by any two of the aforementioned numbers, or about any of the numbers. In some embodiments, the weight ratio of albumin to CSA is about 18:1 or less, such as about 9:1 or less. In some embodiments, the CSA is coated with albumin.

[0186] Alternatively, or in addition, non-biochemical compounds can be added to the pharmaceutical compositions to reduce the toxicity of the therapeutic and/or improve the half-life. Suitable amounts and ratios of an additive that can reduce toxicity can be determined via a cellular assay. With respect to the CSA therapeutic, toxicity reducing compounds can be added to the pharmaceutical composition as 0.25, 0.5, 0.75, 1, 1.5, 2, 2.5, 3, 3.5, 4, 4.5, 5, 10, 20, 50, or 100 weight equivalents, or a range bounded by any two of the aforementioned numbers, or about any of the numbers. In some embodiments, the toxicity reducing compound is a cocoamphodiacetate such as Miranol® (disodium cocoamphodiacetate). In other embodiments, the toxicity reducing compound is an amphoteric surfactant. In some embodiments, the toxicity reducing compound is a surfactant. In other embodiments, the molar ratio of cocoamphodiacetate to CSA is between about 8:1 and 1:1, preferably about 4:1. In some embodiments, the toxicity reducing compound is allantoin.

[0187] In some embodiments, a CSA composition is prepared utilizing one or more surfactants. In specific embodiments, the CSA is complexed with one or more poloxamer surfactants. Poloxamer surfactants are nonionic triblock copolymers composed of a central hydrophobic chain of polyoxypropylene (poly(propylene oxide)) flanked by two hydrophilic chains of polyoxyethylene (poly(ethylene oxide)). In some embodiments, the poloxamer is a liquid, paste, or flake (solid). Examples of suitable poloxamers include those by the trade names Synperonics, Pluronic, or Kolliphor. In some embodiments, one or more of the poloxamer surfactant in the composition is a flake poloxamer. In some embodiments, the one or more poloxamer surfactant in the composition has a molecular weight of about 3600 g/mol for the central hydrophobic chain of polyoxypropylene and has about 70% polyoxyethylene content. In some embodiments, the ratio of the one or more poloxamer to CSA is between about 50 to 1; about 40 to 1; about 30 to 1; about 20 to 1; about 10 to 1; about 5 to 1; about 1 to 1; about 1 to 10; about 1 to 20; about 1 to 30; about 1 to 40; or about 1 to 50. In other embodiments, the ratio of the one or more poloxamer to CSA is between 50 to 1; 40 to 1; 30 to 1; 20 to 1; 10 to 1; 5 to 1; 1 to 1; 1 to 10; 1 to 20; 1 to 30; 1 to 40; or 1 to 50. In some embodiments, the ratio of the one or more poloxamer to CSA is between about 50 to 1 to about 1 to 50. In other embodiments, the ratio of the one or more poloxamer to CSA is between about 30 to 1 to about 3 to 1. In some embodiments, the poloxamer is Pluronic F127.

[0188] The amount of poloxamer may be based upon a weight percentage of the composition. In some embodiments, the amount of poloxamer is about 10%, 15%, 20%, 25%, 30%, 35%, 40%, about any of the aforementioned numbers, or a range bounded by any two of the aforementioned numbers or the formulation. In some embodiments, the one or more poloxamer is between about 10% to about 40% by weight of a formulation administered to the patient. In some embodiments, the one or more poloxamer is between about 20% to about 30% by weight of the formulation. In some embodiments, the formulation contains less than about 50%, 40%, 30%, 20%, 10%, 5%, or 1% of CSA, or about any of the aforementioned numbers. In some embodiments, the formulation contains less than about 20% by weight of CSA.

[0189] The above described poloxamer formulations are particularly suited for the methods of treatment, device coatings, preparation of unit dosage forms (i.e., solutions, mouthwashes, injectables), etc.

[0190] In one embodiment, the compounds described herein may be formulated for oral administration in a lipid-based formulation suitable for low solubility compounds. Lipid-based formulations can generally enhance the oral bioavailability of such compounds.

[0191] A pharmaceutical composition may comprise a therapeutically or prophylactically effective amount of a compound described herein, together with at least one pharmaceutically acceptable excipient selected from the group consisting of medium chain fatty acids or propylene glycol esters thereof (e.g., propylene glycol esters of edible fatty acids such as caprylic and capric fatty acids) and pharmaceutically acceptable surfactants such as polyoxyl 40 hydrogenated castor oil.

[0192] In an alternative embodiment, cyclodextrins may be added as aqueous solubility enhancers. Preferred cyclodextrins include hydroxypropyl, hydroxyethyl, glucosyl, maltosyl and maltotriosyl derivatives of α -, β -, and γ -cyclodextrin. A particularly preferred cyclodextrin solubility enhancer is hydroxypropyl- α -cyclodextrin (BPBC), which may be added to any of the above-described compositions to further improve the aqueous solubility characteristics of the compounds of the embodiments. In one embodiment, the composition comprises about 0.1% to about 20% hydroxypropyl- α -cyclodextrin, more preferably about 1% to about 15% hydroxypropyl- α -cyclodextrin, and even more preferably from about 2.5% to about 10% hydroxypropyl- α -cyclodextrin. The amount of solubility enhancer employed will depend on the amount of the compound of the embodiments in the composition.

[0193] In some exemplary embodiments, a CSA comprises a multimer (e.g., a dimer, trimer, tetramer, or higher order polymer). In some exemplary embodiments, the CSAs can be incorporated into pharmaceutical compositions or formulations. Such pharmaceutical compositions/formulations are useful for administration to a subject, in vivo or ex vivo. Pharmaceutical compositions and formulations include carriers or excipients for administration to a subject.

[0194] Such formulations include solvents (aqueous or non-aqueous), solutions (aqueous or non-aqueous), emulsions (e.g., oil-in-water or water-in-oil), suspensions, syrups, elixirs, dispersion and suspension media, coatings, isotonic and absorption promoting or delaying agents, compatible with pharmaceutical administration or in vivo contact or delivery. Aqueous and non-aqueous solvents, solutions and suspensions may include suspending agents and thickening agents. Such pharmaceutically acceptable carriers include tablets (coated or uncoated), capsules (hard or soft), microbeads, powder, granules and crystals. Supplementary active compounds (e.g., preservatives, antibacterial, antiviral and antifungal agents) can also be incorporated into the compositions.

[0195] Cosolvents and adjuvants may be added to the formulation. Non-limiting examples of cosolvents contain hydroxyl groups or other polar groups, for example, alcohols, such as isopropyl alcohol; glycols, such as propylene glycol, polyethyleneglycol, polypropylene glycol, glycol ether; glycerol; polyoxyethylene alcohols and polyoxyethylene fatty acid esters. Adjuvants include, for example, surfactants such as, soya lecithin and oleic acid; sorbitan esters such as sorbitan trioleate; and polyvinylpyrrolidone.

[0196] A pharmaceutical composition and/or formulation contains a total amount of the active ingredient(s) sufficient to achieve an intended therapeutic effect.

CSA Synthesis

[0197] The methods disclosed herein may be as described below, or by modification of these methods. Ways of modifying the methodology include, among others, temperature, solvent, reagents etc., known to those skilled in the art. In general, during any of the processes for preparation disclosed herein, it may be necessary and/or desirable to protect sensitive or reactive groups on any of the molecules concerned. This may be achieved by means of conventional protecting groups, such as those described in *Protective Groups in Organic Chemistry* (ed. J. F. W. McOmie, Plenum Press, 1973); and P. G. M. Green, T. W. Wutts, *Protecting Groups in Organic Synthesis* (3rd ed.) Wiley, New York (1999), which are both hereby incorporated herein by reference in their entirety. The protecting groups may be removed at a convenient subsequent stage using methods known from the art. Synthetic chemistry transformations useful in synthesizing applicable compounds are known in the art and include e.g. those described in R. Larock, *Comprehensive Organic Transformations*, VCH Publishers, 1989, or L. Paquette, ed., *Encyclopedia*

of *Reagents for Organic Synthesis*, John Wiley and Sons, 1995, which are both hereby incorporated herein by reference in their entirety. The routes shown and described herein are illustrative only and are not intended, nor are they to be construed, to limit the scope of the claims in any manner whatsoever. Those skilled in the art will be able to recognize modifications of the disclosed syntheses and to devise alternate routes based on the disclosures herein; all such modifications and alternate routes are within the scope of the claims.

[0198] Compounds described herein can be prepared by known methods, such as those disclosed in U.S. Pat. No. 6,350,738, which are incorporated herein by reference. A skilled artisan will readily understand that minor variations of starting materials and reagents may be utilized to prepare known and novel cationic steroidal antimicrobials. For example, the preparation of CSA-13 disclosed in U.S. Pat. No. 6,350,738 (compound 133) can be used to prepare CSA-92 by using hexadecylamine rather than octyl amine as disclosed. Schematically, for example, the preparation of certain compounds can be accomplished as follows:

##STR00022##

[0199] As shown above, compound 1-A is converted to the mesylate, compound 1-B using known conditions. Treatment of compound 1-B with a secondary amine, such as HNR.sub.1R.sub.2, results in the formation of compound 1-C, whose azido functional groups are reduced with hydrogen gas in the presence of a suitable catalyst to afford compound 1-D. Suitable catalysts include Palladium on Carbon and Lindlar catalyst. The reagent HNR.sub.1R.sub.2 is not particularly limited under this reaction scheme. For example, when R.sub.1 is hydrogen and R.sub.2 is a C.sub.8-alkyl, CSA-13 is obtained from the synthesis. When R.sub.1 is hydrogen and R.sub.2 is a C.sub.16-alkyl, CSA-92 is obtained from the synthesis. When R.sub.1 and R.sub.2 are both C.sub.5-alkyl, CSA-90 is obtained from the synthesis. A skilled artisan will readily appreciate that this general synthetic scheme can be modified to prepare the CSAs described herein, including CSAs with substituents and functional groups that are different from those generally described above.

[0200] An exemplary but non-limiting general synthetic scheme for preparing compounds of Formula (I), Formula (II), and/or Formula (III) is shown in Scheme B, below. Unless otherwise indicated, the variable definitions are as above for Formulae (I), (II) and/or (III).

##STR00023##

[0201] This process begins with cholic acid (1), or a derivative thereof. Treatment of (1) with a primary or secondary amine R.sub.21R.sub.22NH under amide bond forming conditions yields a final or intermediate CSA compound (2), or a derivative thereof. Amide bond forming conditions include, but are not limited to EDAC [N-(3-dimethylaminopropyl)-N'-ethylcarbodiimide hydrochloride] in the presence of HOBT (1-hydroxybenzotriazole), or HATU [N,N,N',N'-tetramethyl-O-(7-azabenzotriazol-1-yl) uronium hexafluorophosphate) in the presence of diisopropylethylamine, and the like.

[0202] In some embodiments, R.sub.21 and R.sub.22 are independently selected from the group consisting of hydrogen, C.sub.1-C.sub.24 alkyl, C.sub.2-C.sub.24 alkenyl, C.sub.2-C.sub.24 alkynyl, C.sub.6 or C.sub.10 aryl, 5 to 10 membered heteroaryl, 5 to 10 membered heterocyclyl, C.sub.7-13 aralkyl, (5 to 10 membered heteroaryl)—C.sub.1-C.sub.6 alkyl, C.sub.3-10 carbocyclyl, C.sub.4-10 (carbocyclyl)alkyl, (5 to 10 membered heterocyclyl)—C.sub.1-C.sub.6 alkyl, and a suitable amine protecting group, provided that at least one of R.sub.21 or R.sub.22 is not a hydrogen.

[0203] In some embodiments, CSA compound (2), or a derivative thereof, can be treated with an alkoxyacrylonitrile reagent in the presence of acid and a phase transfer catalyst to yield a final or intermediate CSA compound of Formula (3), or a derivative thereof. In some embodiments, the acid is an organic acid. In some embodiments, the acid is an inorganic acid. In some embodiments, the acid is used in catalytic amounts. In some embodiments, the acid is used in stoichiometric amounts. In some embodiments, the acid is used in greater than stoichiometric amounts. In some

embodiments, the phase transfer catalyst is tetrabutylammonium iodide. In some embodiments, the phase transfer catalyst is tetrabutylammonium bromide.

[0204] In some embodiments, CSA Compound (3), or a derivative thereof, can be subjected to reducing conditions suitable for forming CSA compound (4), or a derivative thereof. Suitable reducing conditions include, but are not limited to RedAl , lithium aluminum hydride, lithium borohydride, sodium borohydride, or treatment with hydrogen in the presence of a suitable metal catalyst (e.g., Raney cobalt), or treatment with silyl hydrides in the presence of a suitable metal catalyst. Suitable metal catalysts are known in the art.

[0205] An exemplary synthetic scheme for preparing CSA-192 is shown in Scheme C below.

##STR00024## ##STR00025## ##STR00026##

[0206] In some embodiments, CSA compounds as disclosed herein can be converted into a mesylate salt form, such as to form a pro-drug or hydrolysable intermediate, by reacting one or more amine groups with methylsulfonic acid or derivative thereof (e.g., acid halide). For example, CSA-192 can be converted into its mesylate salt form (CSA-192 MS) by reacting CSA-192 with 3 equivalents of methylsulfonic acid.

EXAMPLES

Counterion Selection

[0207] Counterions were selected based upon toxicity information (i.e., Merck Class 1, 2, and 3), as well as pK_a values, known solubilities of CSA free bases, and the anticipated mode of administration for the drug product.

##STR00027##

[0208] The free base of CSA-13 is obtained by neutralizing the hydrochloride salt as described in U.S. Pat. No. 6,350,738, incorporated herein by reference in its entirety.

pK_a Measurements of CSA-13

[0209] CSA-13 has four basic functional groups. pK_a analysis was performed using the pH-metric method, with the sample being titrated in a triple titration from pH 2.0 to 12.1. CSA-13 pK_a values were measured as 10.77 ± 0.05 , 10.01 ± 0.09 , 9.65 ± 0.04 , and 9.01 ± 0.05 .

Solvent Solubility Test

[0210] Preliminary solubility tests were performed on the free base of CSA-13, reported in Table 1 below:

TABLE-US-00001

TABLE 1	Approximate Solubility	Solvent (mg/mL)	Observations
Acetone	ca. 335		Dissolution was observed. Solvent colour changed to dark brown, after 24 hours at ambient.
Acetonitrile	<10		Initial gum-like material converted to a white solid. After 100 vol., the mixture was cloudy.
1-Butanol	ca. 165		Dissolution was observed.
Cyclohexane	ca. 199		Dissolution was observed.
Dichloromethane	ca. 415		Dissolution was observed.
Diisopropyl ether	<10		Initial gum-like material converted to a white solid. After 100 vol., the mixture was cloudy.
Dimethylformamide	ca. 343		Dissolution was observed.
Dimethylsulfoxide	ca. 246		Dissolution was observed.
1,4-Dioxane	<10		Initial gum-like material converted to a white solid. After 100 vol., the mixture was cloudy.
Ethanol	ca. 406		Dissolution was observed.
Ethyl acetate	<10		Initial gum-like material converted to a white solid. After 100 vol., the mixture was cloudy.
Heptane	<10		Initial gum-like material converted to a white solid. After 100 vol., the mixture was cloudy.
Isopropyl acetate	<10		Initial gum-like material converted to a white solid. After 100 vol., the mixture was cloudy.
Methanol	ca. 400		Dissolution was observed.
Methyl ethyl ketone	ca. 413		Dissolution was observed. Pale yellow after 24 hours at ambient.
Methyl isobutyl ketone	ca. 340		Dissolution was observed. Pale yellow after 24 hours at ambient.
N-Methyl-2-pyrrolidone	ca. 248		Dissolution was observed.
Nitromethane	<10		Complete dissolution was not observed and the colour of the mixture was yellow.
2-Propanol	ca. 263		Dissolution was observed.
tert-Butylmethyl ether	ca. 199		Dissolution was observed.
Tetrahydrofuran	<10		Initial gum-like material converted to a white solid. After 100 vol., the mixture was cloudy.
Toluene	ca. 250		Dissolution was observed.
Water	ca. 205		Dissolution was observed.
Acetonitrile:Water (10%)	ca. 198		Dissolution was observed.

[0211] Solubility values were estimated by a solvent addition technique, based on the following protocol: CSA-13 (20 mg) was weighed and individually distributed to 24 vials. Each solvent was added to the appropriate vial in 10 aliquots of 10 μ L, 5 aliquots of 20 μ L, 3 aliquots of 100 μ L, and 1 aliquot of 500 μ L. If complete dissolution was observed, the additions were stopped. Between additions, the sample was stirred to further encourage dissolution. If 2000 μ L of solvent was added without dissolution, the solubility was calculated to be below this point. Polarized light microscopy analysis was performed on solids obtained from acetonitrile, 1,4-dioxane, ethyl acetate, isopropanol, and THF.

[0212] Based upon the solubility, diversity, toxicity, and stability of CSA-13 in the preliminary solubility tests, the following ICH Class 2 solvents were selected for salt screening experiments: Acetonitrile:Water (10%), Methanol, Tetrahydrofuran, and Toluene. Additionally, 2-Propanol and tert-Butylmethyl ether were also selected.

Counterions for CSA-13 Salt Screening:

[0213] Counterions/acids for the proposed salt screening of CSA-13 were selected on the basis of CSA-13's measured pKa values, described above, and the likelihood of salt formation, which was estimated in part by a greater than about 2 pKa unit difference between the CSA pKa and the free acid pKa of the counterion. Table 2 below lists the counterions/acids identified for preliminary salt screening experiments of CSA-13:

TABLE-US-00002 TABLE 2 Counterion/acid Class pKa 1 pKa 2 pKa 3 Equivalents Benzoic acid 2 4.19 — — 4 Benzenesulphonic acid 2 0.70 — — 2 acid Benzenesulphonic acid 2 0.70 — — 4 acid Citric acid 1 3.13 4.76 6.40 1 Citric acid 1 3.13 4.76 6.40 2 Fumaric acid 1 3.03 4.38 — 2 Galactaric acid 1 3.08 3.63 — 2 (Mucic Acid) Hydrochloric acid 1 -6.10 — — 2 Hydrochloric acid 1 -6.10 — — 4 1-Hydroxy-2- 2 2.70 13.50 — 2 Naphthoic acid 1,5- 2 -3.37 -2.64 — 2 Naphthalenedisulfonic acid Pamoic acid 2 2.51 3.10 — 2 Phosphoric acid 1 1.96 7.12 12.32 4 Succinic acid 1 4.21 5.64 — 2 Sulphuric acid 1 -3.00 1.92 — 2 L-Tartaric acid 1 3.02 4.36 — 2

[0214] Salt screening was carried out using the following protocol: CSA-13 (approximately 25 mg) was slurried or dissolved in the respective solvent, and then mixed with the appropriate equivalents of the acid counterion (specified in Table 2, above). The mixtures of CSA-13/counterion/solvent were temperature cycled between ambient and 40° C. in four hour cycles for a period of approximately 48 hours. The following counterions and solvent combinations were identified from the preliminary screening and advanced to secondary screening:

TABLE-US-00003 TABLE 3 Counterion/acid Equivalent Solvent System 1,5-Naphthalenedisulfonic acid 2 Acetonitrile:water (10%) acid Sulphuric acid 2 tetrahydrofuran Hydrochloric acid 2 tetrahydrofuran Hydrochloric acid 4 tert-Butylmethyl ether Fumaric acid 2 tert-Butylmethyl ether

Secondary Salt Screening: 1,5-Naphthalenedisulfonic Acid

[0215] Approximately 300 mg of CSA-13 was weighed into a scintillation vial. 1.2 mL of acetonitrile:water (10%) was added to the vial. 1,5-Naphthalenedisulfonic acid (2 equivalents) was then added to the vial, resulting in precipitation. A further 1.2 mL of acetonitrile:water (10%) was then added to the vial. The reaction mixture of CSA-13/counterion/solvent was then temperature cycled (40° C./RT, four hour cycles) for approximately 48 hours. Solids were isolated and dried at ambient temperature prior to analysis. Polarized light microscopy of the di-addition salt of 1,5-naphthalenedisulfonate and CSA-13 ("CSA-13 di-NDSA salt") prepared from the secondary salt screening indicated that the material was birefringent and needle-like. FTIR analysis afforded the following results: peaks were identified at about 2925, 2866, 1625, 1500, 1468, 1363, 1240, 1221, 1153, 1108, 1061, 906, 791, 765, 665, 612, 569, 527, and 465 cm⁻¹. The ¹H NMR spectrum for the 1,5-naphthalenedisulfonate di-addition salt of CSA-13 was also obtained. In addition to peaks attributable to the 1,5-naphthalenedisulfonate counterion, shifts in peaks were observed as compared to the free base of CSA-13. HPLC analysis indicated a purity of about 99 percent.

Secondary Salt Screening: Sulfuric Acid

[0216] Approximately 300 mg of CSA-13 was weighed into a scintillation vial. 6 mL of tetrahydrofuran was added to the vial. Sulfuric acid (2 equivalents) was then added to the vial, resulting in slight precipitation. The reaction mixture of CSA-13/counterion/solvent was then temperature cycled (40° C./RT, four hour cycles) for approximately 48 hours. After cycling, a very thin slurry was observed. The solvent was filtered and the solid was dried, affording a gum. The gum was then re-dissolved in 2-propanol, resulting in a slurry that was then temperature cycled (40° C./RT, four hour cycles) for approximately 48 hours. Solids were isolated and dried at ambient temperature prior to analysis.

[0217] Approximately 1 g of CSA-13 was weighed into a scintillation vial. 7 mL of 2-propanol was added to the vial. Sulfuric acid (1 equivalent) was then added to 0.5 mL of 2-propanol, and this solution was added to the vial. The reaction mixture of CSA-13/counterion/solvent was then temperature cycled (40° C./RT, four hour cycles) for approximately 48 hours. After cycling, solvent was evaporated to afford a slurry, which was further temperature cycled (40° C./RT, four hour cycles) for approximately 48 hours. Solids were isolated and analyzed wet by PXRD and then dried at ambient temperature prior to further analysis.

[0218] Analysis of the sulfate salt of CSA-13 prepared from the secondary salt screening indicated that the material was highly crystalline, with no clearly defined morphology. FTIR analysis afforded the following results: peaks were identified at about 2925, 2864, 1618, 1533, 1466, 1364, 1155, 1093, 1027, 854, 611, 579, and 434 cm⁻¹. The ¹H NMR spectrum for the sulfate salt of CSA-13 was also obtained. Shifts in peaks were observed as compared to the free base of CSA-13. HPLC analysis indicated a purity of about 99 percent. Ion chromatography analysis indicates that the ratio of CSA-13 to sulfate counterion was about 1:1.

[0219] A solubility screen was performed as described above for the sulphate salt of CSA-13. The results are provided in Table 4, below:

TABLE-US-00004 TABLE 4 Approximate Solubility Solvent at 22° C. (mg/mL) Acetone <10.5 Acetonitrile <10.4 1-butanol >37.7 Dichloromethane >193.9 1,4-dioxane <11.3 Ethanol >98.4 Methanol >199.7 2-propanol >20.9 TBME <10.1 Tetrahydrofuran <11.0 Toluene >102.8
Secondary Salt Screening: Hydrochloride Salt (2 Equivalents)

[0220] Approximately 300 mg of CSA-13 was weighed into a scintillation vial. 6 mL of tetrahydrofuran was added to the vial. Hydrochloric acid (2 equivalents) was then added to the vial. The reaction mixture of CSA-13/counterion/solvent was then temperature cycled (40° C./RT, four hour cycles) for approximately 48 hours. After cycling, a thin slurry was observed. The solvent was filtered and the solid was dried, affording a gum. The gum was then re-dissolved in 2-propanol, resulting in a slurry that was then temperature cycled (40° C./RT, four hour cycles) for approximately 48 hours. Solids were isolated and dried at ambient temperature prior to analysis. Analysis indicated that the material was not fully crystalline and lacked a defined morphology. Ion chromatography analysis indicated that the ratio of CSA-13 to hydrochloride counterion was about 1:2.5. The material further appeared amorphous after 1 week stability study under all tested conditions.

Secondary Salt Screening: Hydrochloride Salt (4 Equivalents)

[0221] Approximately 300 mg of CSA-13 was weighed into a scintillation vial. 6 mL of tert-butyl methyl ether was added to the vial. Hydrochloric acid (4 equivalents) was then added to the vial. The reaction mixture of CSA-13/counterion/solvent was then temperature cycled (40° C./RT, four hour cycles) for approximately 48 hours. After cycling, heptane anti-solvent addition was performed, resulting in the formation of a gum. The gum was then re-dissolved in 2-propanol and evaporated to afford a solid. The solid was re-slurried in tert-butyl methyl ether and then temperature cycled (40° C./RT, four hour cycles) for approximately 72 hours.

[0222] Analysis indicated that the material was amorphous upon evaporation from the temperature cycle. Further slurrying and temperature cycling for 72 hours failed to afford crystallization.

Additional Salt Screening

Salt No. 1

[0223] Approximately 300 mg of CSA-13 free base was dissolved in 1.5 mL of tert-butylmethyl ether at about 22° C. A sulfuric acid solution was prepared by adding about 1 equivalent (0.44 mmol) of sulfuric acid to 500 µL of tert-butylmethyl ether at about 22° C. The crystallization was seeded using approximately 3-6 mg of seed Form 3. The sulfuric acid solution in tert-butylmethyl ether was added in 50 µL aliquots. The solution was then stirred at about 22° C. for 1 hour. Ethyl acetate (ca. 1.35 mL) was added as an anti-solvent at about 22° C. After anti-solvent addition, the solution was cooled down to 0° C. and the precipitated material is isolated using a centrifuge. The isolated material was dried under vacuum at ambient for 2 hours to provide 285 mg (83% yield) of CSA-13 monosulfate salt as a partially crystalline Form 1 material with 98% purity by HPLC.

Salt No. 2

[0224] Approximately 300 mg of CSA-13 free base was dissolved in 1.5 mL of tert-Butylmethyl ether at about 22° C. A sulfuric acid solution i was prepared by adding about 1 equivalent (0.44 mmol) of sulfuric acid to 500 µL of tert-butylmethyl ether at about 22° C. The crystallization was seeded using approximately 3-6 mg of seed Form 3. The sulfuric acid solution in tert-butylmethyl ether was added in 50 µL aliquots. The solution was then stirred at about 22° C. for 1 hour. The solution was cooled to 5° C. and ethyl acetate (ca. 1.35 mL) was added as an anti-solvent. After anti-solvent addition, the solution was cooled down to 0° C. and the precipitated material was isolated using a centrifuge. The isolated material was dried under vacuum at ambient for 2 hours to provide 248 mg (72% yield) of CSA-13 monosulfate salt as a partially crystalline Form 1 material with 99% purity by HPLC.

Salt No. 3

[0225] Approximately 100 mg of CSA-13 sulfate salt No. 1 was dissolved in 0.75 mL of methanol at ambient (22° C.). The solution was seeded with 1-2 mg of seed (Form 3). About 0.71 mL of ethyl acetate was added and the solution was stirred at about 22° C. for about 1 hour. The solution was cooled down from 22° C. to 5° C. and isolated by centrifugation. The isolated material was dried under vacuum at ambient for 2 hours to provide 90 mg (90% yield) of CSA-13 monosulfate salt as a highly crystalline Form 3 material with 99% purity by HPLC.

Salt No. 4

[0226] Approximately 100 mg of CSA-13 sulfate salt No. 2 was dissolved in 0.75 mL of methanol at ambient (22° C.). The solution was seeded with 1-2 mg of seed (Form 3). About 0.71 mL of ethyl acetate was added and the solution was stirred at about 22° C. for about 1 hour. The solution was cooled down from 22° C. to 5° C. and isolated by centrifugation. The isolated material was dried under vacuum at ambient for 2 hours to provide 86 mg (86% yield) of CSA-13 monosulfate salt as a highly crystalline Form 3 material with 99% purity by HPLC.

Salt No. 5

[0227] Approximately 300 mg of CSA-13 was weighed into a scintillation vial. 6 mL of tert-butyl methyl ether was added to the vial. Fumaric acid (2 equivalents) was then added to the vial. A further 2 mL of tert-butyl methyl ether was added and the reaction mixture of CSA-13/counterion/solvent was then temperature cycled (40° C./RT, four hour cycles) for approximately 48 hours. After cycling, solids were isolated and dried at ambient temperature. PXRD indicated that the material corresponded to fumaric acid. Solids were re-slurried in the mother liquor and then temperature cycled (40° C./RT, four hour cycles) for approximately 72 hours, with the resulting solid determined to be amorphous.

Salt No. 6

[0228] CSA-13 free base was dissolved in EtOH (360 mL) and heated to 60-65° C. A solution of 1,5-naphthalenedisulfonic acid (NDSA) (27.8 g, 77.1 mmol, 2.3 eq) in EtOH/H.sub.2O (1/1 vol/vol; 150 mL) was added over an hour. At the end of the addition, the mixture was cooled to 45° C., seeded (110 mg) and aged overnight at 45° C. The thick slurry obtained was cooled slowly to 0-5° C., held at that temperature for 1-2 hours then isolated by filtration. The cake was washed with

cold EtOH (2×40 mL), dried on the funnel under vacuum and a rubber dam until no further filtrates were observed, then dried in a vacuum oven at 30-40° C. overnight to provide 31.9 g of CSA-13 di-NDSA salt as a white solid.

Salt No. 7

[0229] Approximately 125 mL of ethanol was added to 124 g of CSA-13 free base and the mixture was stirred for 30 minutes at 40° C. for 30 minutes. The mixture was then cooled to 5-10° C.

Separately, 125 mL of ethanol was cooled to 5-10° C. and 11.2 mL of concentrated sulfuric acid was added. The sulfuric acid solution was then added slowly to the CSA-13 free base solution and an exotherm to about 35° C. was observed. The reaction mixture was then stirred at 40° C. for 4 hours. The mixture was allowed to cool overnight to ambient temperature. CSA-13 monosulfate seeds are added and the mixture was cooled to 0-5° and stirred for 4 hours. The mixture was then heated to 40° C. and stirred for 4 hours. The mixture was then allowed to cool overnight to ambient temperature. 1.88 L of MTBE was added to the reaction mixture and the mixture was cooled to 0-5° C. and stirred for 4 hours. The mixture was then heated to 40° C. and stirred for 4 hours. The mixture was then cooled to 0-5° C. and stirred for hours. The reaction mixture was then filtered to obtain 113 g of CSA-13 monosulfate salt with a purity of 97.0% (AUC).

Salt No. 8

[0230] CSA-13 free base (488 mg) was taken up in 10.0 mL of acetonitrile. The mixture was heated to 60-65° C. at which time a solution of 1,5-naphthalenedisulfonic acid (NDSA) (640 mg, 2.5 eq) in 6.0 mL of 1:1 acetonitrile/water was added over about 45 minutes, with solids forming almost immediately (no seeds added). After holding at 60-65° C. for about an hour the batch was slowly cooled to ambient temperature for an overnight stir period. The mixture was cooled in an ice bath and the solids isolated by filtration on a Buchner funnel. After drying (air drying then in a vacuum drying oven), a total of 532 mg of CSA-13 di-NDSA salt was obtained as a pure white solid. The ability of the CSA-13 di-NDSA salt to form a solid precipitate without seeding was surprising and unexpected and shows its unexpectedly superior purification properties compared to acid addition salts of either sulfuric acid or fumaric acid and CSA-13.

Conversion Back to CSA-13 Free Base

[0231] The di-addition salt of NDSA and CSA-13 (0.75 g, 520-068) was combined with 2-MeTHF (7.5 mL), and then an aqueous solution of KOH (0.41 g in 4 mL water) was added. The slurry was aged for 1 h at room temperature during which time a noticeable form change in the slurry was observed. The solids are removed by filtration and the filtrate layers were separated. Toluene (7.5 mL) was added to the organic layer and then washed twice with water (5 mL) before concentrating to an oil to obtain CSA-13 free base (0.5 g). Analysis of the oil and solids indicated no CSA-13 was lost on the solid and that no NDSA remained in the CSA-13 free base. The CSA-13 di-NDSA salt was unexpectedly found to be unique compared to other CSA acid addition salts because none of the CSA-13 was lost when converted back to the free base.

[0232] All x-ray powder diffraction 2θ values are measured with an error of ±0.2 units.

[0233] The CSA-13 monosulfate salt formed herein (as in Salt No. 1 or No. 2) was subjected to XRPD analysis and the pattern shown in FIG. 1 and tabulated in Table 5 were obtained. This material is described as the Form 1 polymorph of the CSA-13 monosulfate salt.

TABLE-US-00005 TABLE 5 Form 1 Peak List Pos. [°2θ] Height [cts]

3.4821	10149.73	4.5781
2575.83	5.2611	3237.31
5.7349	1648.87	7.3569
1698.68	11.5038	2272.18
11.7280	1524.92	13.3929
1827.59	13.9766	1554.22
17.3642	1944.76	17.9760
2308.27	19.0918	2416.90
21.2289	2687.24	

[0234] The CSA-13 monosulfate salt formed as in Salt No. 3 or No. 4 was subjected to XRPD analysis and the pattern shown in FIG. 2 and tabulated in Table 6 were obtained. This material is described as the Form 3 polymorph of the CSA-13 monosulfate salt.

TABLE-US-00006 TABLE 6 Form 3 Peak List Pos. [°2θ] Height [cts]

4.3665	3372.09	4.7145
3615.42	4.9167	11204.68
6.0934	2707.50	6.2547
5888.55	9.4794	4141.07
9.8539	2347.16	10.2449

3408.60 12.8438 6130.97 13.3815 3634.65 14.7948 3394.60 15.9971 1975.64 16.5681 1684.32
18.2047 2482.62 18.3891 2854.19 19.3919 2570.58 20.6269 2699.97 20.8990 2262.26 21.1318
2286.23

[0235] The CSA-13 monosulfate salt prepared as described in Salt No. 5 was subjected to XRPD analysis and the pattern shown in FIG. 3 were obtained, indicating the sample was predominantly amorphous.

[0236] The di-addition salt of NDSA and CSA-13 prepared as in Salt No. 6 was subjected to XRPD analysis and the pattern shown in FIG. 4 and tabulated in Table 7 were obtained.

TABLE-US-00007 TABLE 7 Peak List 2-theta(deg) Height(cps) 4.216(9) 252(29) 4.629(8) 344(34) 8.29(2) 88(17) 9.13(2) 61(14) 9.739(17) 115(20) 12.641(9) 464(39) 14.457(14) 273(30) 15.864(19) 217(27) 18.610(18) 190(25) 19.200(8) 144(22) 20.242(18) 129(21) 20.803(14) 187(25) 21.512(15) 206(26) 22.014(13) 255(29) 22.57(2) 115(20) 23.169(19) 168(24) 23.63(3) 133(21) 25.227(18) 183(25) 26.44(3) 118(20) 37.05(4) 82(16) 39.33(5) 59(14)

[0237] The di-addition salt of NDSA and CSA-13 prepared as in Salt No. 8 was subjected to XRPD analysis and the pattern shown in FIG. 5 and tabulated in Table 8 were obtained.

TABLE-US-00008 TABLE 8 Peak List 2-theta(deg) Height(cps) 4.200(7) 298(31) 4.606(6) 384(36) 8.292(13) 125(20) 9.113(15) 87(17) 9.728(14) 155(23) 11.71(2) 59(14) 12.625(7) 511(41) 13.95(2) 83(17) 14.444(9) 324(33) 15.826(19) 258(29) 18.622(7) 324(33) 19.20(2) 180(24) 20.22(2) 143(22) 20.767(16) 221(27) 21.482(16) 251(29) 21.958(17) 264(30) 22.53(3) 91(17) 23.12(2) 185(25) 23.61(3) 151(22) 25.26(3) 187(25) 26.55(6) 100(18) 37.01(4) 92(17)

[0238] It was unexpectedly found that the formation of the di-addition salt of NDSA and CSA-13 can be used to provide significantly improved purity starting with a less pure form of CSA-13 free base compared to the CSA-13 monosulfate salt. The CSA-13 di-NDSA salt can then be converted back to the free base without the loss of any of the CSA-13 compound. The purified CSA-13 free base can then be converted to CSA-13 monosulfate salt as described herein. Thus, the CSA-13 di-NDSA salt provides an unexpectedly superior and more efficient intermediate when forming the pure CSA-13 monosulfate salt compared to using the CSA-13 monosulfate salt as the intermediate. These unique properties of CSA-13 di-NDSA salt as an intermediate to purification would be expected to be superior other acid addition salts of CSA-13, such as those made using mineral acids, organic acids, and other sulfonic acids.

Summary of Data for CSA-13 Salts:

[0239] The following table summarizes the purity for select CSA-13 salts under various conditions:
TABLE-US-00009 TABLE 9 Salt Conditions Purity (%) 1,5-naphthalenedisulfonate Starting Purity 98.52 (NDSA) salt 40° C./75% RH 97.93 80° C. 98.16 Ambient light 98.57 Sulfate salt Starting Purity 99.34 40° C./75% RH 97.50 80° C. 97.96 Ambient light 99.76

[0240] Based upon the experiments for CSA-13, described above, it was unexpectedly found that the 1,5-naphthalenedisulfonate (NDSA) di-addition salt of CSA-13 had favorable solid state properties and scalability amongst the measured counterions. The CSA-13 monosulfate salt also provided favorable properties, including improved solubility, but did not match the superiority of the CSA-13 di-NDSA salt on ease of crystallization and purification.

##STR00028##

[0241] The free base of CSA-13 can be obtained by neutralizing the hydrochloride salt as described in U.S. Pat. No. 6,350,738, which is incorporated herein by this reference. CSA-131 has some structural similarities with CSA-13. As such, CSA-131 would be expected to have a similar pKa profile. Additionally, it was found that the CSA-131 di-NDSA salt can be prepared, as was the case with CSA-13, and provide similar unexpectedly superior properties of crystallization and purification.

[0242] The free base of CSA-131 (146 g, with an area percent purity of 88.4%) was dissolved in EtOH (2.15 L, 200 proof) and filtered through a 0.20 µm frit into a 5 L reaction flask. The solution was heated to 60-65° C. at which time 1,5-naphthalenedisulfonic acid tetrahydrate (NDSA; 161.5 g,

448 mmol, 2.25 eq.) was added as a solution in 1/1 EtOH/H.sub.2O (900 mL) over 1.75 hours. When approximately 60% of the NDSA solution was added, a small amount of crystallization/precipitation was observed. At the end of the addition of NSA significant solids were present. No seeding was employed. This indicates that the di-addition salt of NDSA and CSA-131 ("CSA-131 di-NDSA salt") was unexpectedly able to crystallize even more readily than the CSA-13 di-NDSA salt.

[0243] The solution was slowly cooled to ambient temperature for an overnight stir period. The next morning the batch was cooled to 0-5° C. and filtered on a funnel to collect the product using ice-cold EtOH to aid in the transfer/provide first rinse of cake (200 mL). The cake was washed with ice-cold EtOH (2×225 mL), dried on the funnel under a latex dam until filtrates ceased, and then dried in a vacuum drying oven until constant weight to provide the CSA-131 di-NDSA salt as a white solid: 197.2 g (75.7% yield) with an HPLC area percent purity of 97.7%.

[0244] A sample of the CSA-131 di-NDSA salt was analyzed by x-ray powder diffraction (XRPD) and the following spectrum was obtained (shown in FIG. 6 and tabulated in Table 10), showing that the salt has a high degree of crystallinity.

TABLE-US-00010 TABLE 10

Pos.	d-spacing	Height	Relative No.	[°2θ]	[Å]	[cts]	Height %
1	4.1922	21.07771	108.88	100.00	2	4.4257	19.96645
2	62.48	57.38	3	6.118	14.44666	5.30	4.87
3	4	8.3931	10.53507	21.73	19.96	5	9.6769
4	9.14015	19.44	17.85	6	11.7232	7.54887	26.83
5	24.64	7	13.4959	6.56107	24.59	22.58	8
6	15.0514	5.88631	59.18	54.35	9	16.5064	5.37059
7	20.03	18.40	10	17.8322	4.97418	54.03	49.62
8	11	18.7671	4.72842	38.17	35.06	12	19.3449
9	4.58848	26.69	24.51	13	20.596	4.31251	46.49
10	42.70	14	21.5538	4.12298	66.44	61.02	15
11	22.7706	3.90535	35.03	32.17	16	24.6057	3.61809
12	30.25	27.78	17	26.7689	3.33041	16.95	15.57
13	18	36.2048	2.48116	5.65	5.19		

[0245] A sample of the CSA-131 di-NDSA salt was subjected to a dynamic vapor sorption (DVS) analysis and results were obtained (FIG. 7), showing that the salt showed minimal hysteresis.

[0246] After being subjected to the DVS analysis, a sample of CSA-131 di-NDSA salt was subjected to XRPD analysis and a spectrum was obtained (shown in FIG. 8 and tabulated in Table 11), showing that the DVS analysis did not significantly impact crystallinity. FIG. 9 provides an overlay of the XRPD spectrum pre- and post-DVS analysis.

TABLE-US-00011 TABLE 11

Pos.	d-spacing	Height	Relative No.	[°2θ]	[Å]	[cts]	Height %
1	4.3296	20.40912	73.81	100.00	2	8.4622	10.44922
2	29.18	39.53	3	9.7475	9.0741	25.19	34.13
3	4	11.8734	7.45376	44.6	60.43	5	13.482
4	6.56779	32.94	44.63	6	15.249	5.81049	57.24
5	77.55	7	16.5541	5.35522	18.92	25.63	8
6	17.8375	4.9727	57.2	77.50	9	18.8803	4.70033
7	46.14	62.51	10	19.4351	4.56739	29.82	40.40
8	11	20.5833	4.31514	47.3	64.08	12	21.5768
9	4.11863	66.11	89.57	13	22.8336	3.8947	41.26
10	55.90	14	24.6093	3.61756	38.51	52.17	15
11	26.8236	3.32374	22.52	30.51	16	32.1213	2.78664
12	2.08	2.82	17	34.323	2.61276	6.29	8.52
13	18	36.2506	2.47813	8.08	10.95		

[0247] Tables 12 and 13 provide the method used to analyze purity of the CSA-131 di-NDSA salt using liquid chromatography with charged aerosol detection (LC-CAD). This method can also be applied to salts of other CSAs, including salts of CSA-13.

TABLE-US-00012 TABLE 12

Column: : Thermo Betasil Phenyl-Hexyl, 50 × 3.0 mm, 3 μm, Part# 73003-053030 Diluent: MeCN/H.sub.2O/TFA (50/50/0.5) Sample Concentration: 1.0 mg/mL for CSA-13 Bis-NDSA Mobile Phase A: Mobile Phase B: MeCN/ H.sub.2O/0.1% TFA MeOH/TFA (80/20/0.1) Column Temperature: Injection Volume: 10 μL 20° C. Sample Temperature: Detection: CAD (Nebulizer: ambient 25° C.; N.sub.2: 35 psi) CAD (Model: ESA Corona, Part# 70-6186A) Gradient Elution Table Time Flow Rate (min) A % B % (mL/min) 0 90 10 1.0 10 54 46 1.0 18 54 46 1.0 20 20 80 1.0 22 20 80 1.0 22.1 90 10 1.0 27 90 10 1.0

TABLE-US-00013 TABLE 13

Method CSA-PHex6D Column Thermo Betasil Phenyl-hexyl 50 × 3 mm, 3 μm Column Temp. 30° C. Detector CAD Mobile phase A: H.sub.2O, 0.1% TFA B: 80% MeCN, 20% MeOH, 0.1% TFA A B Gradient 0.00 min 80 20 10.00 min 15 85 20.00 min 15 85 22.00 min 10 90 23.00 min 80 20 26.00 min 80 20 Flow rate 1.0 mL/min

[0248] Surprisingly it was found that the formation of the CSA-131 di-NDSA salt can be used to

provide significantly improved purity with less pure CSA-131 free base.

##STR00029##

[0249] The free base of CSA-44 can be obtained by neutralizing the hydrochloride salt as described in U.S. Pat. No. 7,598,234, which is incorporated herein by this reference.

pKa Measurements of CSA-44

[0250] CSA-44 has three basic functional groups. pKa analysis was performed using the pH-metric method, with the sample being titrated in a triple titration from pH 2.0 to 12.0. CSA-44 pKa values were measured as 9.15 ± 0.06 , 8.63 ± 0.09 , and 7.75 ± 0.09 .

Solvent Solubility Test

[0251] Preliminary solubility tests were performed on the free base of CSA-44, reported in Table 14 below:

TABLE-US-00014 TABLE 14 Approximate Solubility Solvent (mg/mL) Observations Acetone <10 Initial gum-like material converted to a white solid after 100 μ L. After 100 vol., the mixture was cloudy. Acetonitrile <10 Initial gum-like material converted to a white solid after 100 μ L. After 100 vol., the mixture was cloudy. 1-Butanol <10 Initial gum-like material converted to a white solid after 100 μ L. After 100 vol., the mixture was cloudy. Cyclohexane ca. 104 Dissolution was observed. Dichloromethane ca. 421 Dissolution was observed. Diisopropyl ether <10 Initial gum-like material converted to a white solid after 100 μ L. After 100 vol., the mixture was cloudy. Dimethylformamide ca. 206 Dissolution was observed. Dimethylsulfoxide ca. 208 Dissolution was observed. 1,4-Dioxane <10 Initial gum-like material converted to a white solid after 60 μ L. After 100 vol., the mixture was cloudy. Ethanol <10 Initial gum-like material converted to a white solid after 100 μ L. After 100 vol., the mixture was cloudy. Ethyl acetate <10 Initial gum-like material converted to a white solid after 100 μ L. After 100 vol., the mixture was cloudy. Heptane <10 Dissolution was not observed. Isopropyl acetate <10 Initial gum-like material converted to a white solid after 100 μ L. After 100 vol., the mixture was cloudy. Methanol <10 Initial gum-like material converted to a white solid after 100 μ L. After 100 vol., the mixture was cloudy. Methyl ethyl ketone <10 Initial gum-like material converted to a white solid after 100 μ L. After 100 vol., the mixture was cloudy. Methyl isobutyl ketone <10 Initial gum-like material converted to a white solid after 100 μ L. After 100 vol., the mixture was cloudy. N-Methyl-2-pyrrolidone ca. 107 Dissolution was observed. Nitromethane <10 Initial gum-like material converted to a white solid after 50 μ L. After 100 vol., the mixture was cloudy. 2-Propanol <10 Initial gum-like material converted to a white solid after 100 μ L. After 100 vol., the mixture was cloudy. tert-Butylmethyl ether <10 Initial gum-like material converted to a white solid after 100 μ L. After 100 vol., the mixture was cloudy. Tetrahydrofuran ca. 215 Dissolution was observed. Toluene ca. 424 Dissolution was observed. Water <10 Initial gum-like material converted to a white solid after 100 μ L. After 100 vol., the mixture was cloudy. Acetonitrile:Water (10%) <10 Initial gum-like material converted to a white solid after 200 μ L. After 100 vol., the mixture was cloudy.

[0252] Solubility values were estimated by a solvent addition technique, based on the following protocol: CSA-44 (20 mg) was weighed and individually distributed to 24 vials. Each solvent was added to the appropriate vial in 10 aliquots of 10 μ L, 5 aliquots of 20 μ L, 3 aliquots of 100 μ L, and 1 aliquot of 500 μ L. If complete dissolution was observed, the additions were stopped. Between additions, the sample was stirred to further encourage dissolution. If 2000 μ L of solvent was added without dissolution, the solubility was calculated to be below this point. Polarized light microscopy analysis was performed on solids obtained from acetone, acetonitrile, 1,4-dioxane, ethanol, ethyl acetate, and methanol.

[0253] Based upon the solubility, diversity, toxicity, and stability of CSA-44 in the preliminary solubility tests, the following ICH Class 2 solvents were selected for salt screening experiments: Acetonitrile:Water (10%), Cyclohexane, Tetrahydrofuran, and Toluene. Additionally, 2-Propanol and tert-Butylmethyl ether were also selected.

Counterions for CSA-44 Salt Screening:

[0254] Counterions/acids for the proposed salt screening of CSA-44 were selected on the basis of the measured pKas of CSA-44, described above, and the likelihood of salt formation, which was estimated in part by a greater than about 2 pKa unit difference between the CSA pKa and the free acid pKa of the counterion. Table 15 below lists the counterions identified for preliminary salt screening experiments of CSA-44:

TABLE-US-00015	TABLE 15	Equiv-	Counterion/acid	Class	pKa 1	pKa 2	pKa 3	alents	Benzoic acid
2	4.19	—	—	3	Benzenesulphonic acid	2	0.70	—	—
3	Citric acid	1	3.13	4.76	6.40	1	Citric acid	1	3.13
4.76	6.40	2	Fumaric acid	1	3.03	4.38	—	2	Galactaric acid (Mucic acid)
1	3.08	3.63	—	2	Acid)	Hydrochloric acid	1	—6.10	—
—	2	Hydrochloric acid	1	—6.10	—	—	3	1-Hydroxy-2-Naphthoic acid	2
2.70	13.50	—	3	acid	L-Malic acid	1	3.46	5.10	—
2	1,5-	2	—3.37	—2.64	—	2	Naphthalenedisulphonic acid	Pamoic acid	2
2.51	3.10	—	2	Phosphoric acid	1	1.96	7.12	12.32	3
Succinic acid	1	4.21	5.64	—	2	Sulphuric acid	1	—3.00	1.92
—	2	L-Tartaric acid	1	3.02	4.36	—	2		

[0255] Salt screening was carried out using the following protocol: CSA-44 (approximately 25 mg) was slurried or dissolved in the respective solvent, and then mixed with the appropriate equivalents of the acid counterion (specified in Table 15, above). The mixtures of CSA-44/counterion/solvent were temperature cycled between 5° C. and 25 C in four hour cycles for a period of approximately 48 hours. The following table summarizes the results of the primary salt screen:

TABLE-US-00016	TABLE 16	Solvent	Counterion/acid	Equiv.	A	B	C	D	E	F	Benzoic acid	3	Gum					
AS	PSC	Gum	PSC*	PSC	Benzenesulphonic acid	3	Gum	Gum	PSC*	PSC-	PSC	Gum	Citric acid	1				
Gum	Gum	AS	AS	PSC*	Gum	Citric acid	2	Gum	Gum	AS	AS	PSC*	Gel	Fumaric acid	2	AS	CC	AS
AS	Gum	CC	Mucic acid	2	AS	CC	CC	CC	CC	CC	CC	Hydrochloric acid	2	Gum	Gum	Gum	Gum	Gum
Gum	Hydrochloric acid	3	Gum	Gum	Gum	PSC	Gum	Gum	L-Malic acid	2	Gum	PSC/CC	Gum	AS	Gum	Gum	1,5-Naphthalenedi-	2
Gum	Gum	1,5-Naphthalenedi-	2	PSC*	PSC/CC	PSC	PSC	PSC	PSC/CC	sulphonic acid	Pamoic acid	2	CC	CC	CC	CC	Gum	CC
CC	CC	CC	CC	CC	Gum	CC	Succinic acid	2	Gum	CC	Gum	Gum	PSC*	Gum	1-hydroxy-2-	3	Gum	FB
Gel	PSC*	Gum	PSC*	Naphthoic acid	Phosphoric acid	3	PSC*	PSC*	PSC	Gum	PSC*	Gum	Sulphuric acid	2	Gum	PSC	PSC*	Gum
PSC*	Gum	PSC	PSC*	Gum	PSC*	Gum	L-Tartaric acid	2	Gum	PSC	PSC	PSC	PSC	Gel	L-Aspartic acid	3	Gum	CC
CC	CC	CC	CC	CC	CC	CC	L-Arginine	3	CC	CC	CC	CC	CC	CC	L-tyrosine	3	CC	CC
CC	CC	CC	CC	CC	CC	CC	Meglumine	3	CC	CC	CC	CC	CC	CC	Proline	3	Gum	PSC/CC
PSC/CC	PSC/CC	PSC/CC	PSC/CC	Gum	Urea	3	Gum	Gum	CC	Gum	PSC/CC	Gum	L-Glycine	3	CC/PSC*	CC/PSC*	CC/PSC*	PSC/CC
CC/PSC*	CC/PSC*	PSC/CC	CC/PSC*	PSC/CC	Tromethamine	3	CC	CC	CC	CC	CC	CC	CC	CC	CC	CC	CC	CC

[0256] In Table 16, solvents A-F were as follows: (A) Acetonitrile:water (10%); (B) cyclohexane; (C) 2-propanol; (D) TBME; (E) THF; and (F) toluene. Characterization of the resultant material from the primary screen was as follows: Gum; AS (“amorphous solid”); PSC (“potential salt/co-crystal”); PSC* (“potential salt/co-crystal” obtained with anti-solvent addition); PSC- (“potential salt/co-crystal” obtained by evaporation of solvent); Gel; CC (“counterion/co-former”); and FB (“free base”).

[0257] According to the primary salt screen and provided data, certain samples indicated signs of co-crystal formation. Additional experiments of these samples were performed in which the number of equivalents was reduced from 3 mol to 2 mol and the same salt screening procedure was followed. Isolated material was in the form of a mixture of gum and crystalline solid, with PXRD analysis showing a mixture of PSC and CC.

[0258] Salt screening was also performed using 150 mg of CSA-44, finding that flowable solids could be obtained if material was isolated upon precipitation and without temperature cycling. For experiments resulting in the preparation of thin slurries, it was also found that anti-solvent addition would improve the yield. Amorphous solids were obtained from the following counterions, equivalents, and solvents: Benzoic acid, 3 equivalents, THF; 1,5-naphthalenedisulphonic acid, 2 equivalents, 2-propanol; succinic acid, 2 equivalents, THF; phosphoric acid, 3 equivalents, THF; sulfuric acid, 2 equivalents, TBME; and L-tartaric acid, 2 equivalents, THF. Preliminary results suggested that crystalline material was obtained from the following counterions, equivalents, and solvents: benzenesulfonic acid, 3 equivalents, 2-propanol or THF; and hydrochloric acid, 3

equivalents, TBME. These experiments surprisingly indicated that 1,5-naphthalenedisulphonic acid provided favorable properties such as a stable, flowable solid (from visual inspection).

[0259] To improve crystallinity, amorphous and crystalline solids obtained from the above-described screen were slurried in solvents such as 1,4-dioxane, dichloromethane, methanol, ethyl acetate, diisopropyl ether, and acetonitrile. The results of this experiment are summarized in Table 17:

TABLE-US-00017 TABLE 17 Counterion/Acid 1,4-D DCM M EA DIE ACET Benzoic acid C C A C A CS Benzenesulfonic acid C C C C CS CS Benzenesulfonic acid C CS C C C C Hydrochloric acid CS C C C C CS 1,5-Naphthalenedi- A A C A A A suolfonic acid Succinic acid CS CS CS A A A Phosphoric acid CS A A A A CS Sulfuric acid CS A CS CS CS CS L-Tartaric acid A A A A A A

[0260] In Table 17, 1,4-D stands for “1,4-Dioxane”; DCM stands for “Dichloromethane”; M stands for “Methanol”; EA stands for “Ethyl Acetate”; DIE stands for “Diisopropyl ether”; ACET stands for “Acetonitrole”; C stands for “crystalline”; A stands for “amorphous”; and CS stands for “clear solution.” Although a number of results indicated the formation of crystalline material, 1,5-naphthalenedisulfonic acid (NDSA) appeared to provide the most flowable solid after isolation. Potential salts from benzoic acid showed an improvement in crystallinity in 1,4-dioxane, dichloromethane, and ethyl acetate, but became gum-like upon isolation. Similar results were observed for benzenesulfonic acid and hydrochloric acid. These preliminary data show that the CSA-44 di-NDSA salt was superior to other acid addition salts of CSA-44 relative to crystallization, which would be expected to result in superior purification properties similar to CSA-13 di-NDSA and CSA-131 di-NDSA shown above. Because CSA-44 is structurally different than CSA-13 and CSA-131, the data suggest that di-NDSA salts of other CSAs would similarly be expected to have superior properties relative to crystallization and purification.

Additional Comparative Testing

Example 1

[0261] In a first comparative test, medical devices made from various substrate materials (i.e., stainless steel, titanium, and silicone) were coated with polyurethane coatings containing either di-NDSA salts of CSA compounds (i.e., CSA-13 and/or CSA-131) or HCl acid addition salts of the same CSA compounds, and then tested for long-term stability when immersed in water. Coating stability was determined by measuring how fast the coatings began to sluff off from the medical device. The test results are set forth in Table 18. The NDSA salts are di-addition salts.

TABLE-US-00018 TABLE 18 Substrate Material CSA Acid Sluff Off Time Stainless Steel CSA-13 HCl Within 24 hours Titanium CSA-13 HCl Within 24 hours Silicone CSA-13 HCl Within 24 hours Stainless Steel CSA-131 HCl Within 24 hours Titanium CSA-131 HCl Within 24 hours Silicone CSA-131 HCl Within 24 hours Stainless Steel CSA-13 NDSA None after 30+ days Titanium CSA-13 NDSA None after 30+ days Silicone CSA-13 NDSA None after 30+ days Stainless Steel CSA-131 NDSA None after 30+ days Titanium CSA-131 NDSA None after 30+ days Silicone CSA-131 NDSA None after 30+ days

Example 2

[0262] In a second comparative test, medical devices made from various substrate materials (i.e., stainless steel, titanium, and silicone) were coated with silicone coatings containing either di-NDSA salts of CSA compounds (i.e., CSA-13 and/or CSA-131) or HCl acid addition salts of the same CSA compounds, and then tested for long-term stability when immersed in water. Coating stability was determined by measuring how fast the coatings began to sluff off from the medical device. The test results are set forth in Table 19.

TABLE-US-00019 TABLE 19 Substrate Material CSA Acid Sluff Off Time Stainless Steel CSA-13 HCl 3-4 days Titanium CSA-13 HCl 3-4 days Silicone CSA-13 HCl 3-4 days Stainless Steel CSA-131 HCl 3-4 days Titanium CSA-131 HCl 3-4 days Silicone CSA-131 HCl 3-4 days Stainless Steel CSA-13 NDSA None after 30+ days Titanium CSA-13 NDSA None after 30+ days Silicone CSA-13 NDSA None after 30+ days Stainless Steel CSA-131 NDSA None after 30+ days Titanium

CSA-131 NDSA None after 30+ days Silicone CSA-131 NDSA None after 30+ days

[0263] The foregoing test data in Tables 18 and 19 show that polymer coatings containing di-NDSA salts of CSA-13 and CSA-131 were far more stable and remain adhered to medical devices made from various substrates far longer than polymer coatings containing HCl acid addition salts of the same CSA compounds. The greatly increased stability of coatings made with di-NDSA salts of CSA-13 and CSA-131 yielded medical devices that were suitable for implantation to prevent biofilm formation. By contrast, the low stability of coatings made with HCl acid addition salts of the same CSA compounds yielded medical devices that were unsuitable for implantation to prevent biofilm formation.

Example 3

[0264] In a third comparative test, medical devices made from various substrate materials (i.e., stainless steel, titanium, and silicone) were coated with polyurethane coatings containing either a di-NDSA salt of a representative CSA compound (i.e., CSA-131) or the HCl acid addition salt of the same CSA compound, and then tested to determine the rate of elution of the CSA compound from the coated medical device when immersed in water. Elution rates were determined by measuring the time required for 50% of the CSA compound to elute from the coating. The test results are set forth in Table 20.

TABLE-US-00020 TABLE 20 Time for 50% of Substrate Material CSA Acid CSA to elute
Stainless Steel CSA-131 HCl Within 2 days Titanium CSA-131 HCl Within 2 days Silicone CSA-131 HCl Within 2 days
Stainless Steel CSA-131 NDSA >7 days Titanium CSA-131 NDSA >7 days
Silicone CSA-131 NDSA >7 days

Example 4

[0265] In a fourth comparative test, medical devices made from various substrate materials (i.e., stainless steel, titanium, and silicone) were coated with silicone coatings containing either a di-NDSA salt of a representative CSA compound (i.e., CSA-131) or the HCl acid addition salt of the same CSA compound, and then tested to determine the rate of elution of the CSA compound from the coated medical device when immersed in water. Elution rates were determined by measuring the time required for 50% of the CSA compound to elute from the coating. The test results are set forth in Table 21.

TABLE-US-00021 TABLE 21 Time for 50% of Substrate Material CSA Acid CSA to elute
Stainless Steel CSA-131 HCl Within 2 days Titanium CSA-131 HCl Within 2 days Silicone CSA-131 HCl Within 2 days
Stainless Steel CSA-131 NDSA >7 days Titanium CSA-131 NDSA >7 days
Silicone CSA-131 NDSA >7 days

Example 5

[0266] In a fifth comparative test, medical devices were made from PLGA copolymer impregnated with either a di-NDSA salt of a representative CSA compound (i.e., CSA-131) or the HCl acid addition salt of the same CSA compound, and then tested to determine the rate of elution of the CSA compound from the medical device when immersed in water. Elution rates were determined by measuring the time required for 50% of the CSA compound to elute from the medical device. The test results are set forth in Table 22.

TABLE-US-00022 TABLE 22 Time for 50% of Polymer Material CSA Acid CSA to elute PLGA
CSA-131 HCl Within 2 days PLGA CSA-131 NDSA >7 days

[0267] The foregoing test data in Tables 20-22 show that the di-NDSA salt of CSA-131 provided for significantly more controlled and longer elution of CSA compounds compared to the HCl acid addition salts of the same CSA compound. The controlled and longer release of the CSA compound when making coatings and medical devices from polymers containing the di-NDSA salt of CSA-131 was significantly more effective in preventing biofilm formation and maintaining a sterile outer surface of the medical devices compared to coatings and medical devices made from polymers containing the HCl acid addition salt of the same CSA compound.

Example 6

[0268] In a sixth comparative test, the ability of silicone polymers to cross-link using platinum (Pt) catalysts when using different acid addition salts of CSA compounds was compared. Including HCl acid addition salts of CSA-13 and CSA-131 in silicone polymers greatly inhibited cross-linking. More specifically, HCl acid addition salts of the CSA compounds quenched Pt catalyst activity in silicone coatings and inhibited cross-linking. By contrast, Pt catalysts remained active when including di-NDSA salts of the same CSA compounds in silicone polymers. Thus, the di-NDSA salts of CSA compounds did not inhibit cross-linking of silicone polymers. This differential in cross-linking ability greatly improved the adhesion and stability of silicone polymer coatings containing di-NDSA salts of CSA compounds compared to silicone polymer coatings containing HCl acid addition salts of the same CSA compounds.

Example 7

[0269] In a seventh comparative test, the elution rate and antibacterial activity of polyurethane coatings containing either the di-NDSA salt of a representative CSA compound (i.e., CSA-131) or the HCl acid addition salt of the same CSA compound following sterilization with ethylene oxide (EtO) were compared, and also the tendency of the acid addition salt of the CSA compound to react with the EtO.

[0270] For EtO reactivity with the HCl acid addition salt of CSA-131, the following percentages of CSA-131 and CSA-131-EtO adducts were observed via mass spectroscopy following EtO treatment of a medical device with a polyurethane coating containing the HCl acid addition salt of CSA-131. The test results are shown in Table 23.

TABLE-US-00023 TABLE 23 CSA-131: 68% CSA-131 bonded to one molecule of EtO: 20% CSA-131 bonded to two molecules of EtO: 3% CSA-131 bonded to three molecules of EtO: 2%

[0271] The elution rate of CSA-131 from the same polyurethane coating containing the HCl acid addition salt of CSA-131 (with solution exchange every 24 hours) is shown in Table 24:

TABLE-US-00024 TABLE 24 24 hours: 16.0 µg 48 hours: 4.2 µg 72 hours: undetectable

[0272] The low amounts of CSA-13 that eluted from the polyurethane coating that contained the HCl acid addition salt of CSA-131 (as compared to the amount of CSA-13 that eluted from the same polyurethane coating, but which contained the di-NDSA salt of CSA-131, as discussed below) appear to be the result of undesired reactions of the HCl acid addition salt of CSA-131 with the polyurethane coating.

[0273] Because there was no detectable elution of CSA-131 from the polyurethane coating containing the HCl acid addition salt of CSA-13 at 72 hours, this coating failed to prevent bacterial growth (MRSA or Pa01) on Day 3.

[0274] By comparison, even after an endotracheal tube (ETT) with a polyurethane coating containing the di-NDSA salt of CSA-131 was sterilized multiple times with EtO, no EtO adducts were observed by mass spectroscopy. Moreover, elution of CSA-131 from the ETT with the same polyurethane coating containing the di-NDSA of CSA-131 remained relatively consistent for 96 hours, providing continuous activity against biofilms. This is shown by the data in Table 25.

TABLE-US-00025 TABLE 25 Time Interval Average Amount .sup.1Eluted Amount in .sup.2% of (hours) Eluted (µg) µg/cm.sup.3 Total µg on OD 0-24 94.3 0.95 2.37% 24-48 84.9 0.85 2.14% 48-72 64.9 0.65 1.63% 72-96 80.1 0.81 2.01% Total 324 µg 3.26 µg/cm.sup.3 8.15% .sup.1Total OD Surface Area (assumes 28 cm of coated length) - 99.4 cm.sup.2 .sup.2Total CSA-131 Content on OD (@ 40 µg/cm.sup.2) - 3976 µg

[0275] Moreover, the polyurethane coating containing the di-NDSA salt of CSA-131 prevented bacterial and fungal growth for 4 to 16 days, as shown in the graphs depicted in FIG. 10, which was in sharp contrast to the failure of the polyurethane coating containing the HCl acid addition salt of the same CSA compound to prevent bacterial growth, as stated above.

[0276] FIG. 10 includes five graphs, labeled (a)-(e), that depict the amount of biofilm growth on endotracheal tube (ETT) 5 mm segments that were either uncoated (black bars) or coated with polyurethane containing the di-NDSA salt of CSA-131 (grey bars). The bars show the amount of

biofilm growth on the ETT 5 mm segments after the indicated number of days of incubation (daily exchange of growth medium and re-inoculation). Measurements were made in triplicate. Graph (a) compares biofilm growth of MRSA (BAA-41, 10.sup.6 cfu inoculation). Graph (b) compares biofilm growth of *K. pneumoniae* (ATCC 13883, 10.sup.6 cfu inoculation). Graph (c) compares biofilm growth of *P. aeruginosa* (ATCC 47085, 10.sup.6 cfu inoculation). Graph (d) compares biofilm growth of *C. albicans* (ATCC 90028, 10.sup.6 cfu inoculation). Graph (e) compares biofilm growth of *C. auris* (CDC 0383, 10.sup.6 cfu inoculation).

Example 8

[0277] In an eighth comparative test, acid addition salts were made from a representative CSA compound (i.e., CSA-131) and either 1,5-naphthalenedisulfonic acid (NDSA) or 2-naphthalenesulfonic acid (2-NSA) to form respective CSA-131 acid addition salts. Because NDSA has two sulfonic acid groups and the di-addition salt includes two moles of NDSA per mole of CSA-131, the di-addition salt of NDSA and CSA-131 (“CSA-131 di-NDSA salt”) includes four sulfonic acid groups that form ionic bonds with the four amine groups of CSA-131. Because 2-NSA only has one sulfonic acid group, the comparative test utilized the tetra-addition salt (“CSA-131 tetra-NSA salt”) in order for the acid addition salt of 2-NSA and CSA-131 to include four sulfonic acid groups that form ionic bonds with the four amine groups of CSA-131. This ensured that the test was a valid comparison of the properties of the CSA-131 acid addition salts of NDSA and 2-NSA.

[0278] The solubilities of the CSA-131 acid addition salts of NDSA and 2-NSA were tested in either water or ethanol, a polar organic solvent, in a manner similar to the solvent solubility tests described above in previous examples (see tables above listing solubilities in the various solvents). The CSA-131 di-NDSA salt was unexpectedly found to have much lower solubility in both water and ethanol, while the CSA-131 tetra-NSA salt unexpectedly had significantly higher solubilities in water and ethanol. It is hypothesized that that 1,5-naphthalenedisulfonic acid creates a bridging effect between two amine groups of the CSA-131 that greatly affects solubility. The solubilities of the two acid additional salts are set forth in Table 26:

TABLE-US-00026 TABLE 26 Solubility Solubility CSA Salt in Ethanol in Water CSA-131 di-NDSA <50 µg/mL <20 µg/mL CSA 131 tetra-NSA >5 mg/mL ~100 µg/mL

[0279] The very low solubility of CSA-131 di-NDSA in ethanol makes it a useful acid addition salt for isolation and purification of CSA-131 di-NDSA because it is readily crystallized, which facilitates easy purification. In contrast, the relatively high solubility of CSA 131 tetra-NSA makes it a poor acid addition salt for isolation and purification, as crystallization would either be impossible or unfavorably low, which forecloses this form of purification. The unexpectedly lower solubility of CSA-131 di-NDSA compared to CSA 131 tetra-NSA in ethanol is what provides these divergent results and highlights an important benefit of the surprising and unexpectedly superior properties of the CSA-131 di-NDSA salt compared to the CSA-131 tetra-NSA salt.

[0280] With respect to their different solubilities in water, the very low solubility of CSA-131 di-NDSA in water has been found to be extremely useful for controlling the rate of elution of CSA-131 from medical device coatings. In contrast, the significantly greater (>5-fold) solubility of CSA 131 tetra-NSA in water results in a significantly higher elution rate of CSA-131 from medical device coatings. The unexpectedly lower solubility of CSA-131 di-NDSA compared to CSA 131 tetra-NSA in water is what provides these divergent results and highlights another important benefit of the surprising and unexpectedly superior properties of the CSA-131 di-NDSA salt compared to the CSA-131 tetra-NSA salt.

[0281] Based on the solubility testing, the CSA-131 di-NDSA salt is able to form coatings for medical devices, such as endotracheal tubes, that have significantly greater stability compared to coatings made using the CSA-131 tetra-NSA salt, with coatings containing the CSA-131 di-NDSA salt being expected to exhibit significantly more controlled and sustained elution of CSA-131 compared to coatings containing the CSA-131 tetra-NSA salt.

[0282] In summary, the Examples in this disclosure set forth detailed test methods comparing a wide variety of acid addition salts of CSA compounds (e.g., CSA-13, CSA-44 and CSA-131), including di-NDSA addition salts, sulfuric acid addition salts, fumaric acid addition salts, HCl acid addition salts, and 2-NSA addition salts. The test data demonstrate that di-NDSA salts of various CSA compounds were more easily purified by precipitation of salt form, followed by removal of the acid and conversion of the CSA compounds back into their free base form, compared to other acid addition salts, including sulfuric acid addition salts, fumaric acid addition salts, HCl acid addition salts, and 2-NSA addition salts, of the same or similar CSA compounds.

[0283] In addition, the test data in this disclosure show that di-NDSA addition salts of CSA compounds can be more easily formed into crystalline solids compared to other acid addition salts, including sulfuric acid addition salts, fumaric acid addition salts, HCl acid addition salts, and 2-NSA addition salts, of the same or similar CSA compounds. This greatly increases ease of purification of the di-NDSA salts of CSA compounds compared to the other acid addition salts of the same or similar CSA compounds.

[0284] In addition, the test data in this disclosure show that di-NDSA salts of CSA compounds can be more easily formed into stable, flowable solids compared to other acid addition salts, including sulfuric acid addition salts, fumaric acid addition salts, and HCl acid addition salts of the same or similar CSA compounds.

[0285] In summary, the test data set forth in this disclosure clearly demonstrate that di-NDSA addition salts of CSA compounds were unexpectedly and unpredictably superior to addition salts of other acids, including sulfuric acid addition salts, fumaric acid addition salts, HCl acid addition salts, and 2-NSA addition salts of the same or similar CSA compounds, when tested relative to coating stability (comparative tests 1 and 2 in Examples 1 and 2), elution rate (comparative tests 3, 4 and 5 in Examples 3-5), ability of silicone polymers to crosslink (comparative test 6 in Example 6), ability to elute and provide antimicrobial activity following sterilization with EtO (comparative test 7 in Example 8), and ease of purification and controlled rate of elution (e.g., comparative test 8 in Example 8).

CONCLUSION

[0286] The present invention may be embodied in other specific forms without departing from its spirit or essential characteristics. The described embodiments are to be considered in all respects only as illustrative and not restrictive. The scope of the invention is, therefore, indicated by the appended claims rather than by the foregoing description. All changes which come within the meaning and range of equivalency of the claims are to be embraced within their scope.

Claims

1. A di-addition salt of 1,5-naphthalene disulfonic acid (NDSA) and a cationic steroidal antimicrobial (CSA) compound of Formula III: ##STR00030## wherein, R.sub.3, R.sub.7, and R.sub.12, are each an unsubstituted (C.sub.1-C.sub.18) aminoalkyloxy; and R.sub.18 is selected from the group consisting of unsubstituted (C.sub.1-C.sub.18) alkylamino-(C.sub.1-C.sub.18) alkyl and unsubstituted di(C.sub.1-C.sub.18 alkyl)aminoalkyl.
2. The di-addition salt of claim 1, wherein: R.sub.3, R.sub.7, and R.sub.12 are each unsubstituted (C.sub.1-C.sub.5) aminoalkyloxy; and R.sub.18 is selected from the group consisting of unsubstituted (C.sub.1-C.sub.16) alkylamino-(C.sub.1-C.sub.5) alkyl and unsubstituted di(C.sub.1-C.sub.5alkyl)amino-(C.sub.1-C.sub.5) alkyl.
3. The di-addition salt of claim 1, wherein: R.sub.3, R.sub.7, and R.sub.12 are amino-C.sub.3-alkyloxy; and R.sub.18 is selected from the group consisting of C.sub.8-alkylamino-C.sub.5-alkyl; C.sub.12-alkylamino-C.sub.5-alkyl; C.sub.13-alkylamino-C.sub.5-alkyl; C.sub.16-alkylamino-C.sub.5-alkyl; and di-(C.sub.5-alkyl)amino-C.sub.5-alkyl.
4. The di-addition salt of claim 1, wherein: R.sub.3, R.sub.7, and R.sub.12 are amino-C.sub.3-

alkyloxy, and R.sub.18 is selected from the group consisting of C.sub.8-alkylamino-C.sub.5-alkyl; C.sub.12-alkylamino-C.sub.5-alkyl; C.sub.13-alkylamino-C.sub.5-alkyl; and C.sub.16-alkylamino-C.sub.5-alkyl.

5. The di-addition salt of claim 4, wherein R.sub.18 is selected from the group consisting of C.sub.12-alkylamino-C.sub.5-alkyl and C.sub.8-alkylamino-C.sub.5-alkyl.

6. The di-addition salt of claim 1, wherein the CSA compound is selected from the group consisting of: ##STR00031##

7. The di-addition salt of claim 1, wherein the di-addition salt is a solid.

8. The di-addition salt of claim 7, wherein the solid is a flowable solid.

9. The di-addition salt of claim 1, wherein the di-addition salt is crystalline.

10. The di-addition salt of claim 1, wherein the di-addition salt is storage stable.

11. The di-addition salt of claim 1, wherein the di-addition salt is micronized.

12. The di-addition salt of claim 1, wherein the di-addition salt is characterized by an x-ray powder diffraction pattern with the following 2θ values (± 0.2): 4.216; 4.629; 8.29; 9.13; 9.739; 12.641; 14.457; 15.864; 18.610; 19.200; 20.242; 20.803; 21.512; 22.014; 22.57; 23.169; 23.63; 25.227; 26.44; 37.05; and 39.33.

13. The di-addition salt of claim 1, wherein the di-addition salt is characterized by an x-ray powder diffraction pattern with the following 2θ values (± 0.2): 4.200; 4.606; 8.292; 9.113; 9.728; 11.71; 12.625; 13.95; 14.444; 15.826; 18.622; 19.20; 20.22; 20.767; 21.482; 21.958; 22.53; 23.12; 23.61; 25.26; 26.55; and 37.01.

14. A formulation, comprising the di-addition salt of claim 1 and a pharmaceutically acceptable excipient.

15. A di-addition salt of 1,5-naphthalenedisulfonic acid (NDSA) and a cationic steroidal antimicrobial (CSA) compound of Formula III, which is a crystalline solid: ##STR00032## wherein, R.sub.3, R.sub.7, and R.sub.12, are each an unsubstituted (C.sub.1-C.sub.22) aminoalkyloxy; and R.sub.18 is selected from the group consisting of (C.sub.1-C.sub.22) alkylamino-(C.sub.1-C.sub.22) alkyl and di(C.sub.1-C.sub.22 alkyl)aminoalkyl.

16. The di-addition salt of claim 15, wherein the CSA is selected from the group consisting of: ##STR00033##

17. A di-addition salt of 1,5-naphthalenedisulfonic acid (NDSA) and a cationic steroidal antimicrobial (CSA) compound of Formula III, which is a flowable solid: ##STR00034## wherein R.sub.3, R.sub.7, and R.sub.12, are each an unsubstituted (C.sub.1-C.sub.22) aminoalkyloxy; and R.sub.18 is (C.sub.1-C.sub.22) alkylamino-(C.sub.1-C.sub.22) alkyl.

18. A process for preparing the di-addition salt of NDSA and a CSA compound of claim 1, comprising: diluting the free base of the CSA compound with a solvent; adding at least two equivalents of 1,5-naphthalenedisulfonic acid (NDSA) to the diluted CSA compound in the solvent to yield a reaction mixture; precipitating or temperature cycling the reaction mixture; and isolating the di-addition salt of NDSA and the CSA compound.

19. The process of claim 18, wherein the temperature cycling is conducted for at least about 48 hours.

20. The process of claim 18, further comprising utilizing an anti-solvent or evaporation of solvent when isolating the di-addition salt of NDSA and the CSA compound.
